

Mannich reaction-based combinatorial libraries identify antioxidant ionizable lipids for mRNA delivery with reduced immunogenicity

Received: 6 April 2024

Accepted: 8 May 2025

Published online: 18 July 2025

 Check for updates

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The immunogenicity of lipid nanoparticles (LNPs) used for the delivery of nucleoside-modified messenger RNA limits the levels and durability of expression of the encoded protein. Here, by leveraging the Mannich reaction for ionizable lipid synthesis, and via the *in vitro* and *in vivo* screening of six combinatorial libraries of synthesized lipids, we report the identification of an antioxidant ionizable lipid, C-a16, exhibiting reduced immunogenicity. When incorporated into LNPs for mRNA delivery, C-a16 mitigated the generation of intracellular reactive oxygen species, thereby extending the duration of protein expression. In mice, and compared with commercial LNPs, LNPs incorporating C-a16 and co-delivering Cas9 mRNA and guide RNA for the editing of the transthyretin gene led to 2.8-fold higher editing efficiency; LNPs with C-a16 delivering fibroblast growth factor 21 mRNA increased the expression of the protein 3.6-fold; and when delivering mRNA encoding a tumour neoantigen or the spike protein of SARS-CoV-2, LNPs with C-a16 induced stronger antigen-specific immune responses. Our findings support the further testing of C-a16 as a promising ionizable lipid for mRNA delivery in therapeutic applications.

Nucleoside-modified messenger ribonucleic acid (mRNA)-based therapeutics have revolutionized disease treatment^{1,2}. A pioneering study demonstrated that introducing pseudouridine modifications to mRNA species can substantially reduce their immunogenicity and enhance their stability within cells, resulting in improved intracellular protein expression³. Lipid nanoparticles (LNPs)⁴ serve as an optimal vehicle for delivering mRNA-based therapeutics owing to their exceptional efficacy and improved safety profiles compared with viral-based delivery systems⁵. LNPs have recently demonstrated their clinical efficacy, with notable examples including an LNP formulation that delivers

small interfering RNA (siRNA) for hereditary transthyretin-mediated amyloidosis (Onpattro)⁶, as well as the Pfizer-BioNTech and Moderna coronavirus disease 2019 (COVID-19) mRNA vaccines^{7,8}. Nevertheless, the broader applications of existing LNP-based delivery systems are limited owing to their intrinsic high immunogenicity^{9,10}. Studies have shown that LNPs can trigger innate immune responses in cells¹⁰, diminishing the potential therapeutic benefits of nucleoside-modified mRNA. This immunogenicity challenge is further exemplified by the observed side effects associated with the Pfizer-BioNTech and Moderna COVID-19 mRNA vaccines⁹ following intramuscular injection, as well as the severe

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systemic inflammatory syndrome observed upon systemic injection of Onpatro^{6,11}. These studies demonstrate the urgent need for the development of mRNA delivery systems with lower immunogenicity to enhance therapeutic efficacy and reduce side effects¹².

Combinatorial chemistry-based high-throughput screening strategies enable efficient identification of potent ionizable lipids from hundreds or thousands of candidates within days, and thus are a powerful strategy for lipid discovery^{13,14}. These combinatorial chemistries are usually aminomethylation reactions, including the broadly used amine-acrylate/acrylamide Michael addition reactions¹⁵ and amine-epoxide ring-opening reactions^{15,16}. Even though the current Michael addition- and epoxide-based combinatorial chemistries can generate lipids with diverse structures, the types of substrate are limited and thus cannot meet the increasing demand for ionizable lipids. Mannich reactions are a potentially promising alternative. Mannich aminomethylation^{17,18} consists of the condensation of a substrate possessing at least one active hydrogen with formaldehyde and a primary or secondary amine. The substrates of Mannich reactions can contain many different chemical groups, including phenols, thiols, alkyl ketones, carboxylic acid derivatives and heterocyclic compounds (including pyrrole, furan and thiophene)¹⁹. Mannich reactions can generate compounds with secondary and tertiary amines with high yields¹⁷ and thus could lead to the creation of ionizable lipid libraries with greater structural and chemical diversity.

In this study, we designed six unique three-component combinatorial libraries, utilizing reactions involving amines and formaldehyde with a range of compounds, including phenols, thiols, dialkyl ketones, pyrroles and acetophenones. Following a comprehensive screening of 310 lipids *in vitro* and 11 lipids *in vivo*, we identified an ionizable lipid, C-a16, with antioxidant properties ideal for mRNA delivery with low immunogenicity. We assessed the efficacy of C-a16 LNPs for multiple applications, including delivering Cas9 mRNA and single guide RNA (sgRNA) targeting transthyretin (*Ttr*) for genome editing, fibroblast growth factor 21 (FGF21) for protein supplementation, tumour neoantigen for cancer vaccination and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) spike protein for COVID-19 vaccination. We used commercially available DLin-MC3-DMA and SM-102 lipids in LNP formulations as benchmarks. Our results demonstrate the potential of the C-a16 ionizable lipid for achieving mRNA delivery with reduced immunogenicity across a range of therapeutic applications.

Results and discussion

Mannich lipid library construction and screening

The Mannich reaction is a three-component condensation in which a compound containing an active hydrogen atom is reacted with formaldehyde and an NH-amine derivative (Fig. 1a). We designed a series of Mannich reaction libraries utilizing a range of starting materials, including phenols, thiols, alkyl ketones, pyrroles and acetophenones (Supplementary Fig. 1). Different tails were numbered alphabetically as shown in Fig. 1b,c and Supplementary Figs. 4b, 7b, 10b, 13b and 15b. For each type of Mannich reaction, the derivatives were mixed with formaldehyde and amines at a specific ratio in ethanol alone or ethanol and water (Supplementary Figs. 1–16). These mixtures were then

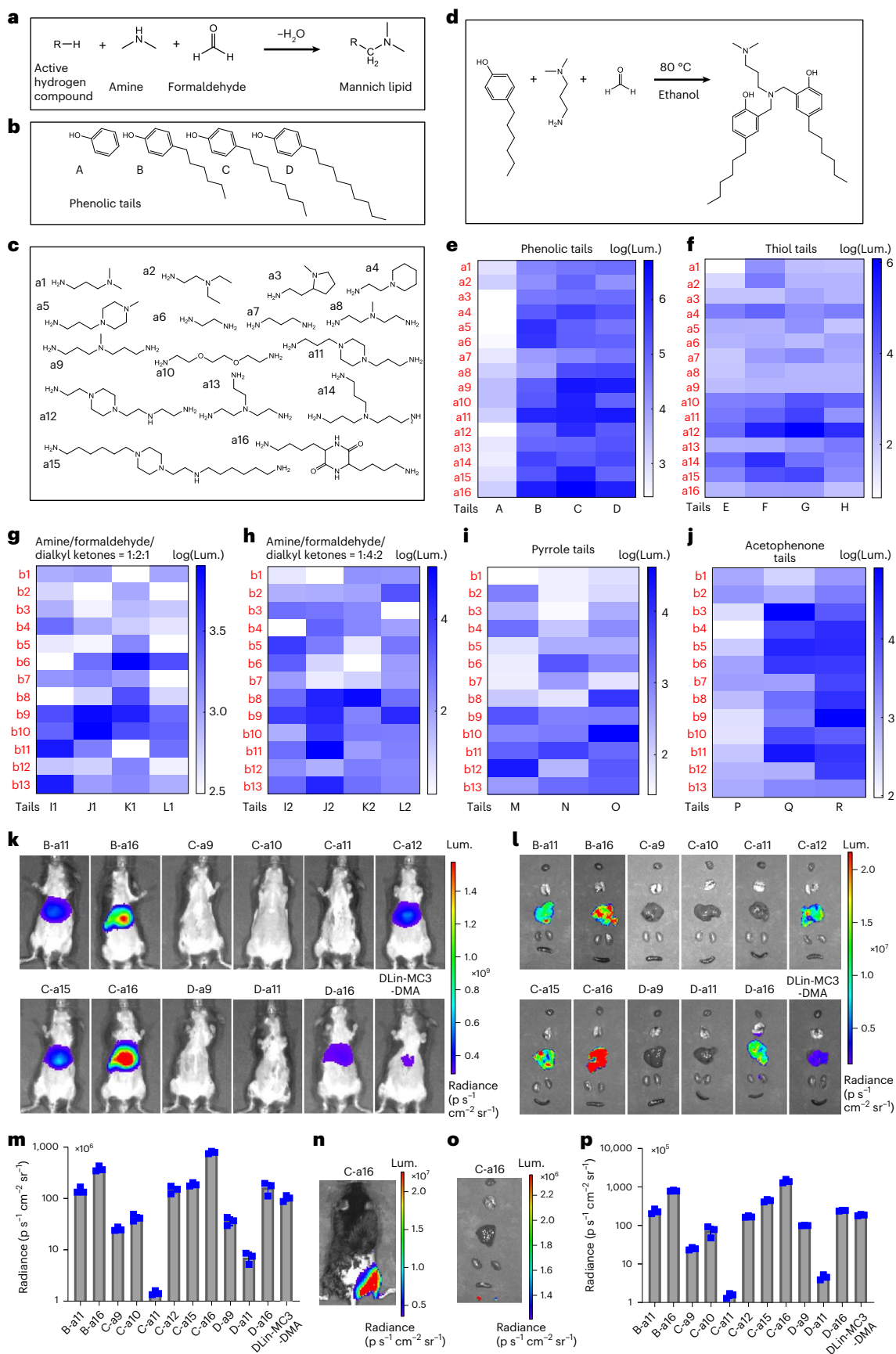
heated to 80 °C for 1–24 h depending on the reaction and are specified in Supplementary Figs. 1, 4, 7, 10, 13 and 15. The synthesis of B-a1 lipid is exemplified in Fig. 1d. After the reactions were completed, the solvents were evaporated to obtain 310 different crude lipid compounds. These crude compounds were subsequently utilized to prepare LNPs for *in vitro* screening.

We prepared LNPs encapsulating luciferase-encoding mRNA (5-methoxyuridine-modified) through a pipette mixing process. This involved mixing an ethanol phase containing crude ionizable lipid, 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (DMG-PEG 2000) and cholesterol, with a citrate buffer phase containing mRNA. A mouse colon cancer cell line MC38 was used for *in vitro* screening experiments. The cells were treated with LNPs at a dose of 10 ng per well mRNA. Twenty-four hours later, luciferase assay substrate was added and luminescence was measured. Notably, lipids synthesized using the phenol Mannich reaction exhibited higher luminescence levels when compared with lipids derived from other Mannich reactions (Fig. 1e–j). We also pooled all the lipids of each reaction type together, formulated LNPs and treated MC38 cells, and found that once again the phenol reaction lipids had the highest luminescence (Supplementary Fig. 17a). Moreover, for a specific amine, *N,N*-diethylethylenediamine (referred to as amine a2 in phenol- and thiol-based libraries and b10 in alkyl ketone-, carboxylic acid derivative-, pyrrole- and acetophenone-based libraries), we found that the B-a2 lipid with phenolic tails induced a significant increase in transfection compared with other lipids with similar carbon numbers on the tail ends but generated from other Mannich libraries (Supplementary Fig. 17b). This suggests the potentially unique role of the phenol group in these lipids. Among all the lipids in the phenol Mannich library, we found that phenol A, lacking a carbon chain on the benzene ring, failed to transfect the cells. By contrast, phenols with alkyl chains on the benzene ring consistently exhibited relatively higher transfection. We defined LNPs with luminescence levels exceeding five times that of the commercialized DLin-MC3-DMA lipid as ‘hits’. Our results indicated that lipids synthesized with phenols A, B, C and D resulted in 0, 3, 6 and 2 ‘hits’, respectively. This demonstrates that phenol C, featuring an 8-carbon alkyl group on the benzene ring, possesses superior transfection efficiency compared with other phenols (Supplementary Fig. 17c). Notably, while all LNPs showed similar size, polydispersity index (PDI), encapsulation efficiency (Supplementary Fig. 17d) and cytotoxicity (Supplementary Fig. 17e), the C-a16 lipid was identified as the top performer, demonstrating ~15-fold higher luminescence intensity compared with the DLin-MC3-DMA LNP formulation (Supplementary Fig. 17f).

Next, we tested the lipids that met the ‘hit’ criteria for *in vivo* mRNA delivery in mice. LNPs encapsulating luciferase mRNA were intravenously injected into mice at a dose of 0.125 mg kg^{−1} mRNA. After 6 h, the expression of luciferase *in vivo* was examined. Our results showed that the luciferase signal was mainly detected in the liver for all LNPs, with the C-a16 LNP showing the highest luminescence intensity (Fig. 1k,l). Notably, the luminescence level of the isolated liver in C-a16 LNP-treated mice was approximately fourfold higher than that of

Fig. 1 | Synthesis and evaluation of six combinatorial libraries of Mannich ionizable lipids for mRNA delivery. **a**, The Mannich reaction is a three-component condensation in which a compound containing an active hydrogen atom is reacted with formaldehyde and an NH-amine derivative. **b–d**, Synthesis of the phenolic tail-based ionizable lipid library: phenolic tails (**b**) and amine cores (**c**) used for the phenolic tail-based combinatorial library and exemplified synthesis of B-a1 lipid (**d**). **e–j**, Heat maps showing the luminescence (Lum.) levels of MC38 cells following treatment with luciferase mRNA encapsulated in various LNPs formulated using phenolic tail-based lipids (**e**), thiol tail-based lipids (**f**), dialkyl ketone tail-based lipids synthesized with dialkyl ketone:formaldehyde:amine ratio of 1:2:1 (**g**), dialkyl ketone tail-based lipids

synthesized with dialkyl ketone:formaldehyde:amine ratio of 1:4:2 (**h**), pyrrole tail-based lipids (**i**) and acetophenone tail-based lipids (**j**). **k,l**, Intravenous injection of the top 11 performing LNPs identified in **e–j** and subsequent assessment of their biodistribution in mice (**k**) and various organs including heart, lung, liver, kidney and spleen (**l**). **m**, Quantification of luminescence levels in livers in **l** ($n = 3$ biological replicates). **n,o**, *In vivo* luminescence signal visualization in mice (**n**) and different organs (**o**) through IVIS after intramuscular injection of C-a16 LNPs. **p**, Quantification of the luminescence levels in mice following intramuscular injection of the top-performing 11 LNPs. Data in **m** and **p** are shown as mean $\pm$ s.d. ($n = 3$ biological replicates).



DLin-MC3-DMA LNPs (Fig. 1m), highlighting its potential for systemic delivery applications. In addition, we examined LNP performance after intramuscular injection. Mice were injected with LNPs encapsulating luciferase mRNA at a dose of 0.125 mg kg⁻¹ mRNA, and the luminescence level at the injection site was assessed 6 h post-injection. We found that C-a16 LNP not only induced the highest luciferase expression at the injection site (Fig. 1n–p and Supplementary Fig. 18a) but also led to the highest luminescence expression in the lymph nodes (Supplementary Fig. 18b,c). Lymph node delivery is required for vaccine applications, as is the transfection of antigen-presenting cells such as dendritic cells and macrophages to trigger cellular and humoral immune responses towards antigens. Thus, these results suggest that the C-a16 LNP also holds promise as a formulation for vaccine applications.

C-a16 LNP improves mRNA delivery by quenching ROS

We purified the C-a16 lipid and characterized it using mass spectrometry and ¹H NMR (Supplementary Fig. 19a,b). We also characterized C-a16 LNPs. Transmission electron microscopy and dynamic light scattering showed that C-a16 LNPs had comparable morphology, size distribution and size stability over time to DLin-MC3-DMA LNPs (Fig. 2a–c and Supplementary Fig. 20a–h). In addition, we evaluated luciferase expression kinetics in mice injected with DLin-MC3-DMA and C-a16 LNPs (Fig. 2d). The half-life of luciferase signal in DLin-MC3-DMA LNP-treated mice was approximately 22 h (Fig. 2d), whereas in C-a16 LNP-treated mice, it extended to around 26 h (Fig. 2d). This roughly 18.2% increase in expression durability suggests that degradation of the luciferase mRNA potentially occurs at a slower rate in the C-a16 LNP group compared with the DLin-MC3-DMA LNP group. Subsequently, we compared the C-a16 lipid with different commercially available ionizable lipids, such as DLin-MC3-DMA, SM-102 and ALC-0315, with regard to their expression durability of luciferase and GFP expression (Supplementary Fig. 20i,j). Our results demonstrated that C-a16 LNPs led to prolonged luciferase and GFP expression in cells (Supplementary Fig. 20i,j), while the cellular uptake of various LNPs remained similar (Supplementary Fig. 21). Altogether, our results demonstrate that C-a16 LNPs facilitate durable mRNA expression compared with LNPs made with commercially available ionizable lipids.

We then wanted to explore the mechanism underlying the sustained mRNA expression achieved with C-a16 LNPs. Given that the C-a16 lipid contains four phenol groups, we hypothesized that these phenol groups could contribute to the high transfection efficiency and prolonged expression durability. To investigate this, we chemically modified C-a16 to produce a C-a16-Q lipid by quenching its phenol groups using acetyl chloride (Supplementary Fig. 22a). Our results showed a substantial reduction in both in vitro and in vivo transfection efficiency for C-a16-Q LNPs when compared with C-a16 LNPs (Supplementary Fig. 22b–d). In addition, we assessed the durability of luciferase expression in mice treated with C-a16-Q LNPs, and observed a significant decrease compared with mice treated with C-a16 LNPs (Fig. 2d). We further investigated whether the C-a16 LNPs induce changes in innate immune activation. C-a16, C-a16-Q or DLin-MC3-DMA LNPs were intravenously administered into mice and serum interleukin-6 (IL-6) levels were measured 24 h post-LNP injection. We found that C-a16 LNP treatment resulted in decreased IL-6 secretion in mouse blood

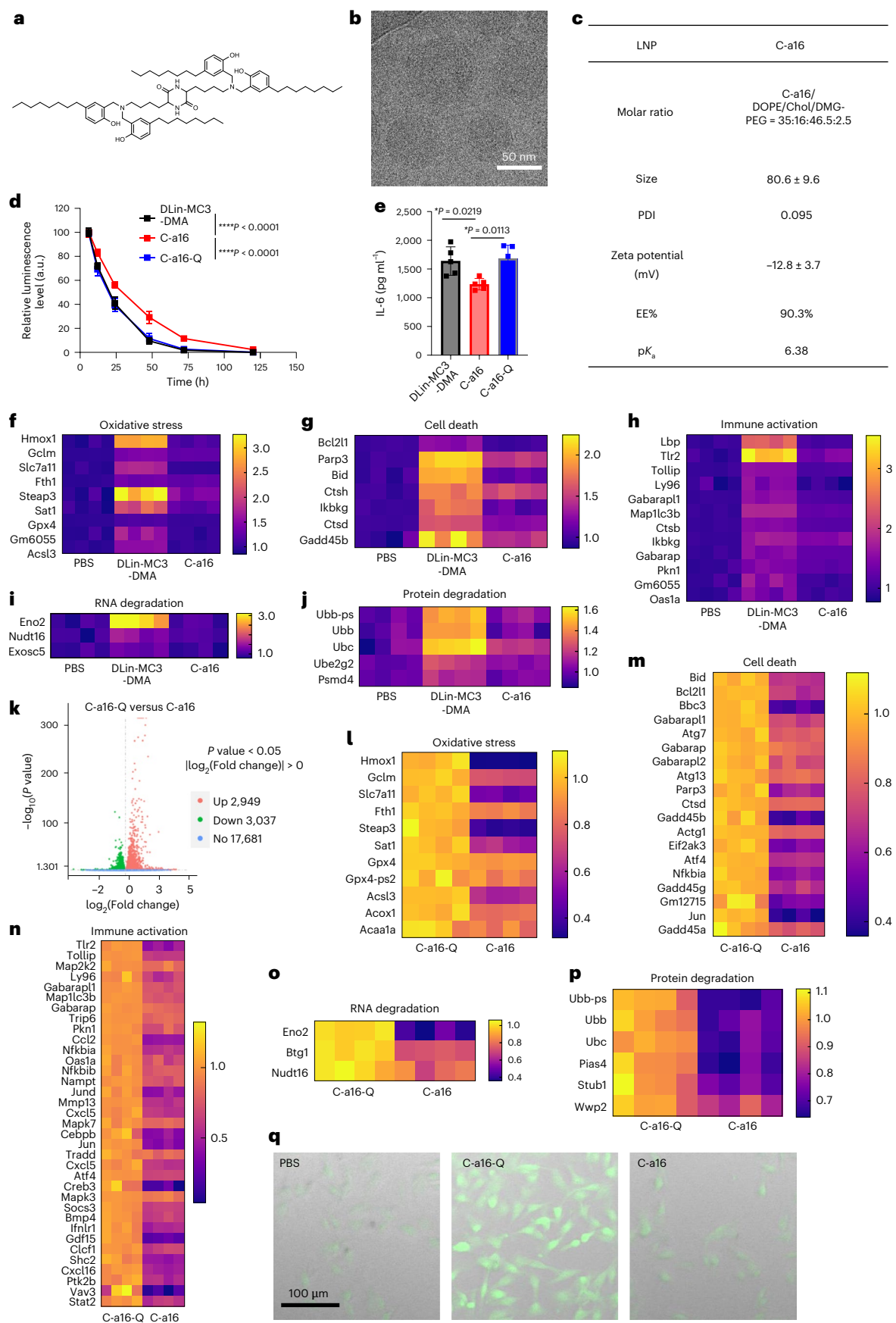
compared with C-a16-Q and DLin-MC3-DMA LNPs (Fig. 2e). Moreover, C-a16-Q and DLin-MC3-DMA LNPs induced significant increases in the number of immune cells, including neutrophils, monocytes and lymphocytes, indicating activation of inflammatory responses (Extended Data Fig. 1a–d). By contrast, the numbers of neutrophils, monocytes and lymphocytes in the C-a16 LNP-treated group were comparable to those in the PBS-treated group, suggesting that C-a16 LNPs did not trigger innate immune activation. In addition, C-a16-Q and DLin-MC3-DMA LNPs led to higher levels of tumour necrosis factor (TNF)-α and IL-2 in the blood, compared with C-a16 LNPs (Extended Data Fig. 1e,f), further demonstrating that the phenol groups in C-a16 LNP induce limited innate immune activation. These results collectively emphasize the critical role played by the phenol groups in enhancing transfection efficiency and sustaining mRNA expression delivered by C-a16 LNPs.

Studies have shown that polyphenol compounds^{20–22} possess antioxidant and anti-inflammatory properties owing to the phenol groups that they contain, which can interact with various free radicals²⁰. To investigate the mechanism of C-a16 LNP-mediated mRNA delivery, an RNA-sequencing experiment was performed. Since the proteins produced by mRNA translation can also activate innate immune responses, potentially influencing the assessment of the impact of the delivery carriers (LNPs) themselves on cells, we examined the kinetics of LNP uptake and mRNA expression in vitro (Supplementary Fig. 23). We observed that the uptake of LNPs by MC38 cells gradually increased over time, reaching a plateau at 3 h post-LNP treatment (Supplementary Fig. 23a). mRNA translation began around 4 h post-LNP treatment and reached a plateau at approximately 6 h post-LNP treatment (Supplementary Fig. 23b,c). As a result, we chose the 3 h time point precluding mRNA translation to evaluate the effects of the LNPs on cells. MC38 cells were treated with PBS, DLin-MC3-DMA LNPs or C-a16 LNPs for 3 h, after which the cells were collected, and total RNA was extracted. Through RNA sequencing (Fig. 2f–j), we found that DLin-MC3-DMA LNPs induced elevated intracellular oxidative stress, characterized by the upregulation of genes such as *Hmox1*, *Gclm*, *Slc7a11*, *Fth1*, *Steap3*, *Sat1*, *Gpx4*, *Gm6055* and *AcsL3* (Fig. 2f). In addition, pathways related to cell death and innate immune activation were also greatly enhanced (Fig. 2g,h). Furthermore, we found that DLin-MC3-DMA LNP treatment resulted in increased expression of genes related to RNA and protein degradation (Fig. 2i,j), which is consistent with previous research indicating that intracellular oxidative stress²³ and innate immune activation hampers RNA translation and triggers RNA and protein degradation^{24–26}. By contrast, the expression of oxidative stress, cell death, innate immune activation, and RNA- and protein-degradation genes in C-a16 LNP-treated cells was much lower than in cells treated with DLin-MC3-DMA LNPs (Fig. 2f–j). The contrast in oxidative stress and innate immune activation induced by DLin-MC3-DMA LNPs and C-a16 LNPs was further confirmed through confocal and enzyme-linked immunosorbent assay (ELISA) experiments (Supplementary Fig. 24a–c).

We further conducted experiments to investigate the degradation of mRNA delivered by C-a16 LNPs, C-a16-Q LNPs and DLin-MC3-DMA LNPs, as well as degradation of protein produced by the translation of the mRNA. DLin-MC3-DMA LNPs, C-a16 LNPs and C-a16-Q LNPs were formulated, and mRNA encoding for the tumour neoantigen

Fig. 2 | Potent and durable mRNA transfection by C-a16 LNPs. **a**, C-a16 lipid structure. **b**, Cryo-EM image of C-a16 LNPs. **c**, Table depicting the formulation characteristics of C-a16 LNPs, including size, PDI, zeta potential, encapsulation efficiency (EE%) and pK_a. **d**, Total luminescence levels in mice treated with DLin-MC3-DMA, C-a16 or C-a16-Q LNPs at various time points (6 h, 12 h, 24 h, 48 h, 72 h and 120 h). Luminescence signals at each time point were normalized to the signal at 6 h. **e**, Measurement of serum IL-6 levels 24 h after LNP injection. Data in **d** and **e** are shown as mean ± s.d. (*n* = 5 biological replicates). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test. **f–j**, Heat map of differentially expressed genes (DEGs) associated with oxidative stress (**f**), cell

death (**g**), immune activation (**h**), RNA degradation (**i**) and protein degradation (**j**) in response to PBS, DLin-MC3-DMA LNP or C-a16 LNP treatment. **k**, Volcano plots highlighting DEGs after treatment with C-a16 LNP and C-a16-Q LNP compared with PBS treatment. Statistical differences were calculated using a Wald test. **l–p**, Heat map illustrating selected DEGs associated with oxidative stress (**l**), cell death (**m**), immune activation (**n**), RNA degradation (**o**) and protein degradation (**p**) in response to treatment with C-a16 LNP or C-a16-Q LNP (*n* = 4 biological replicates). **q**, Cells were treated with PBS, C-a16-Q LNPs and C-a16 LNPs for 3 h, and the intracellular ROS levels were determined using a DCFH-DA dye. For **b** and **q**, experiments were repeated three times independently with similar results.



PDZ-binding kinase (Pbk)-actin alpha 4 (Actn4) was used as a model mRNA. A FLAG tag was added to obtain Pbk-Actn4-FLAG for easier evaluation of the protein degradation inside cells. MC38 cells were treated with DLin-MC3-DMA, C-a16 or C-a16-Q LNPs encapsulating mRNA encoding Pbk-Actn4-FLAG. Afterwards, the cells were collected at 6 h or 24 h and intracellular mRNA and proteins were extracted. Through a real-time quantitative PCR experiment, we demonstrated that the level of intracellular Pbk-Actn4-FLAG mRNA was similar in DLin-MC3-DMA LNP-, C-a16 LNP- and C-a16-Q LNP-treated cells at 6 h post-LNP treatment (Supplementary Fig. 25a). However, the level of Pbk-Actn4-FLAG mRNA in C-a16-Q LNP- and DLin-MC3-DMA LNP-treated cells greatly decreased 24 h post-LNP treatment, with less of a decrease in C-a16 LNP-treated cells at 24 h post-LNP treatment (Supplementary Fig. 25a). These results demonstrate reduced degradation of Pbk-Actn4-FLAG mRNA when delivered by C-a16 LNPs compared with that delivered by C-a16-Q LNPs and DLin-MC3-DMA LNPs.

We also evaluated if C-a16 LNPs affect protein degradation levels inside cells. As ubiquitination of proteins and subsequent degradation by proteasomes is a major mechanism for intracellular protein degradation, we conducted a western blot assay to evaluate if C-a16 LNPs can reduce the ubiquitination of intracellular proteins compared with C-a16-Q LNPs and DLin-MC3-DMA LNPs. Ubiquitination levels were similar among C-a16-, C-a16-Q- and DLin-MC3-DMA-treated groups 6 h post-LNP treatment. However, while C-a16-Q LNPs and DLin-MC3-DMA LNPs led to a substantial increase in ubiquitination of intracellular proteins 24 h post-LNP treatment, the level of ubiquitination of intracellular proteins in C-a16 LNP-treated cells was much lower (Supplementary Fig. 25b). Moreover, we found that the level of Pbk-Actn4-FLAG protein in cells that received DLin-MC3-DMA LNPs, C-a16 LNPs and C-a16-Q LNPs was similar at 6 h post-LNP treatment, indicating similar protein translation levels. However, the level of Pbk-Actn4-FLAG in C-a16-Q LNP- and DLin-MC3-DMA LNP-treated cells was much lower 24 h post-LNP treatment, compared with that in C-a16 LNP-treated cells (Supplementary Fig. 25c). These results are consistent with previous reports that higher reactive oxygen species (ROS) levels inside cells lead to protein degradation^{27,28}. Altogether, these results demonstrate that C-a16 LNP treatment decreases mRNA and protein degradation inside cells, therefore improving the duration of target protein expression. Furthermore, elevated oxidative stress induced by H₂O₂ treatment and increased innate immunity activation by lipopolysaccharide treatment led to decreased luciferase expression (Supplementary Fig. 26a,b). Altogether, these results suggest that the superior mRNA delivery ability of C-a16 LNPs is attributed to the reduced activation of oxidative stress- and innate immunity-related pathways.

We also investigated whether the reduced activation of cellular oxidative stress and innate immune responses is a result of the phenol groups in the C-a16 lipid structure. We analysed changes in gene expression in cells treated with C-a16 LNPs and C-a16-Q LNPs. Our results demonstrated that, following the quenching of the phenol groups in the C-a16 lipid, C-a16-Q LNPs triggered elevated levels of genes associated with oxidative stress, cell death, innate immune activation, and RNA and protein degradation (Fig. 2k–p). In addition, we observed that quenching the phenol groups in the C-a16 lipid resulted

in higher levels of ROS within the cells (Fig. 2q). We performed further experiments to investigate whether phenol groups could inhibit ROS production and increase mRNA expression durability in other ionizable lipids. We quenched the phenol groups in several lipids, including B-a11, C-a15, B-a16, D-a9, C-a12 and D-a16, to obtain B-a11-Q, C-a15-Q, B-a16-Q, D-a9-Q, C-a12-Q and D-a16-Q, respectively, and formulated LNPs with these lipids and treated cells for 6 h. We found that quenching of phenol hydroxyl groups in these lipids resulted in a substantial increase in intracellular ROS levels across these lipids (Supplementary Fig. 27a). Moreover, quenching of phenol groups in these lipids led to substantial inhibition of mRNA expression (Supplementary Fig. 27b). These results suggest that the introduction of phenol groups into lipids to mitigate ROS production and increase mRNA expression applies to a broad range of ionizable lipids. These results further demonstrate the critical role of phenol groups in mRNA delivery.

C-a16 LNP for gene editor delivery in vivo

Motivated by the promising results of our in vitro experiments, we assessed the potential of C-a16 LNPs for in vivo mRNA delivery. We assessed the in vivo biodistribution of C-a16 LNPs in a gene editing mouse model. DLin-MC3-DMA, C-a16 and C-a16-Q LNPs encapsulating mRNA encoding Cre recombinase were intravenously injected into Ai9 mice at a dose of 0.3 mg kg⁻¹ mRNA. Ai9 mice have a *loxP*-flanked STOP cassette, which prevents transcription of tdTomato, and thus efficient LNP delivery leads to Cre-mediated recombination and tdTomato expression²⁹. Notably, we observed that C-a16 LNPs induced higher levels of tdTomato expression in the liver compared with DLin-MC3-DMA LNPs and C-a16-Q LNPs. The tdTomato signal was mainly detected in hepatocytes, F4/80⁺ Kupffer cells and CD31⁺ endothelial cells (Fig. 3a). Furthermore, we performed flow cytometry to investigate LNP biodistribution in the liver (Fig. 3b–e and Supplementary Fig. 28a,b). While DLin-MC3-DMA LNPs resulted in tdTomato expression in approximately 38% of hepatocytes, 23% of Kupffer cells and 31% of endothelial cells, C-a16 LNPs led to substantially higher expression levels, with 69%, 81% and 62% expression in each respective cell subset in the liver (Fig. 3c–e). We also evaluated ROS levels in liver cells collected from mice treated with LNPs (Supplementary Fig. 29). The liver cell ROS levels in mice treated with C-a16 LNPs were significantly lower than those in the DLin-MC3-DMA LNP-treated group (Supplementary Fig. 29). Quenching the phenol group in the C-a16 lipid reversed this effect (Supplementary Fig. 29). Altogether, these results suggest that the antioxidative properties of the C-a16 lipid contribute to the high transfection efficiency of C-a16 LNPs.

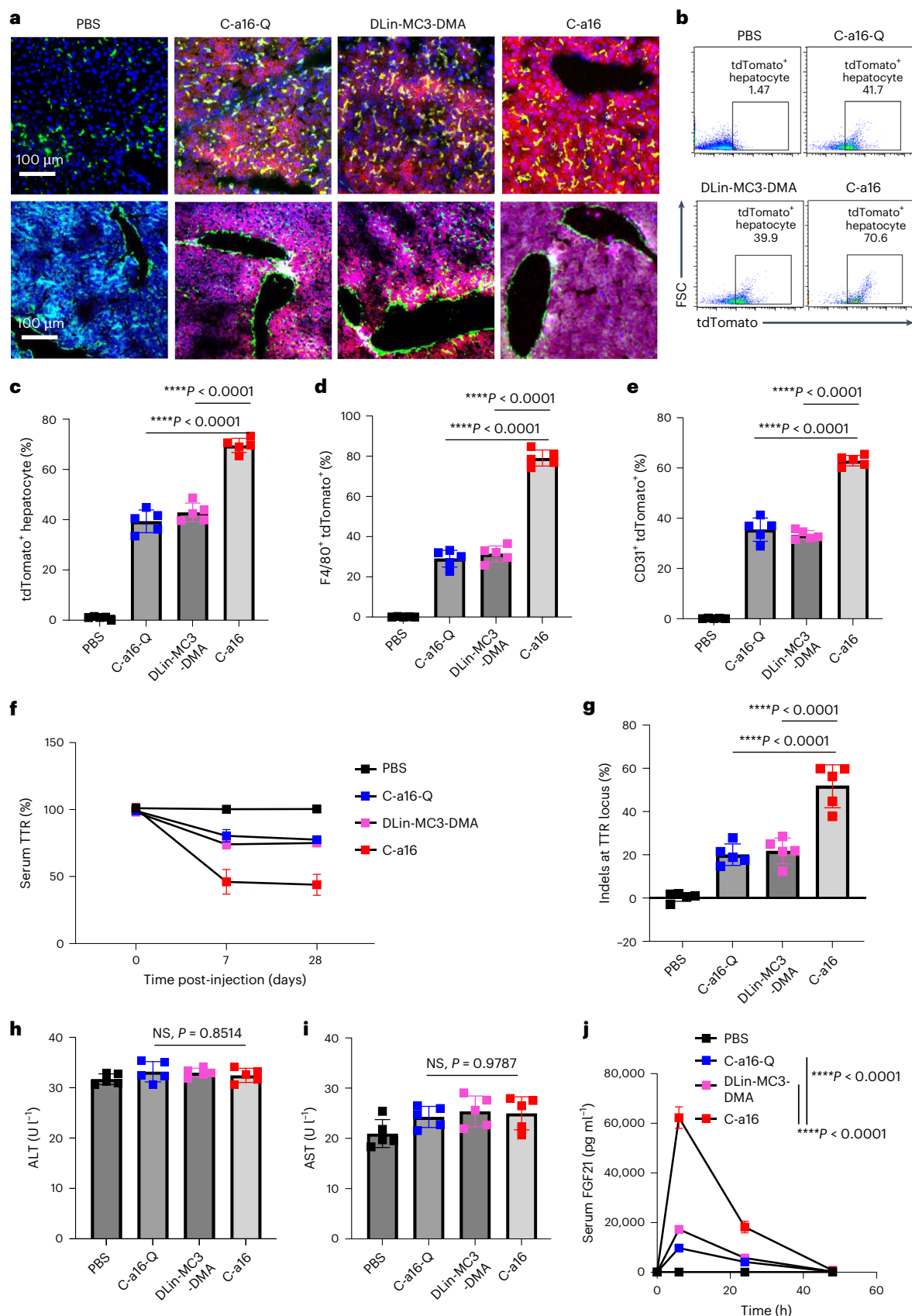
Next, we evaluated the C-a16 LNP for functional mRNA delivery. Given the encouraging results seen with Intellia's LNP-mediated liver transthyretin (*Ttr*) knockout using the CRISPR–Cas9 system^{30,31}, we aimed to explore the potential of our C-a16 LNPs in the context of *Ttr* gene editing. A single injection of C-a16 LNPs, co-encapsulating Cas9 mRNA and a *Ttr* sgRNA, at a total RNA dose of 2 mg kg⁻¹ resulted in an ~55% reduction in serum *Ttr* levels and a 50% rate of insertions and deletions (indels) at the *Ttr* gene locus (Fig. 3f,g). By contrast, DLin-MC3-DMA LNPs achieved only a 20% reduction in serum TTR levels and an approximately 18% indel rate at the *Ttr* locus (Fig. 3f,g).

Fig. 3 | Systemic mRNA delivery using C-a16 LNPs. a–e, LNPs delivering mRNA encoding Cre recombinase were intravenously injected into Ai9 mice on day 0. Mice were euthanized on day 3, and the livers were subjected to an immunofluorescence assay (**a**). Liver Kupffer cells and endothelial cells were labelled with F4/80 and CD31 antibodies, respectively (green, upper panels for F4/80, lower panels for CD31). The distribution of LNPs within various liver cell types was assessed by observing tdTomato fluorescence using confocal microscopy. Biodistribution of different LNPs in liver cell populations was further investigated using flow cytometry (**b**), and the percentage of tdTomato⁺ hepatocytes (**c**), tdTomato⁺ Kupffer cells (**d**) and tdTomato⁺ endothelial cells (**e**) was quantified. **f,g**, LNP-mediated co-delivery of Cas9 mRNA and *Ttr* sgRNA for

transthyretin (*Ttr*) gene editing. Mice received intravenous injections of LNPs co-delivering Cas9 mRNA and *Ttr* sgRNA (4:1, wt/wt) at a total RNA dose of 2 mg kg⁻¹. Serum samples collected at specified time points were analysed for serum TTR levels by ELISA (**f**). On day 7, DNA extracted from the livers was used to determine on-target indel frequency by next-generation sequencing (**g**). **h,i**, Serum levels of liver enzymes ALT (**h**) and AST (**i**) 24 h post-LNP injection. **j**, LNP-mediated delivery of FGF21 mRNA. Mice received intravenous injections of LNPs encapsulating FGF21 mRNA at an mRNA dose of 1 mg kg⁻¹. Serum samples collected at different time points were analysed for FGF21 levels by ELISA. Data in **c–j** are presented as mean ± s.d. (*n* = 5 biological replicates). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test. NS, not significant.

As expected, C-a16-Q LNPs induced a much lower reduction in serum TTR levels and indels at the *Ttr* locus in mice compared with those induced by C-a16 LNPs (Fig. 3f,g). Furthermore, the levels of alanine transaminase (ALT) and aspartate aminotransferase (AST)

in LNP-treated mice remained within normal ranges (Fig. 3h,i), indicating that C-a16 LNP treatment did not induce liver toxicity. These results demonstrate the potential of C-a16 LNPs for systemic delivery of mRNA-based gene editors.



Next, we assessed the delivery of therapeutic mRNA encoding human FGF21 using C-a16 LNPs. FGF21 is a secreted endocrine factor that regulates metabolism, promoting the oxidation of fatty acids and the production of ketone bodies and inhibiting lipogenesis^{32,33}. FGF21 supplementation is a promising treatment for a range of metabolic disorders, including obesity, diabetes, mitochondrial diseases and ageing^{34,35}. Our results showed that in mice treated with C-a16 LNPs encapsulating FGF21 mRNA (1 mg kg^{-1}), a notable increase in circulating FGF21 was observed, reaching $62,204 \text{ pg ml}^{-1}$ in the blood 6 h post-LNP injection (Fig. 3j). This concentration was 3.6-fold and 6.4-fold higher than in mice treated with DLin-MC3-DMA LNPs ($17,204 \text{ pg ml}^{-1}$) and C-a16-Q LNPs ($9,717.2 \text{ pg ml}^{-1}$), respectively. In addition, we calculated the area under the curve for FGF21 expression over time. The area under the curve in C-a16 LNP-treated mice was 3.46-fold and 5.56-fold greater than that in mice treated with DLin-MC3-DMA LNPs and C-a16-Q LNPs, respectively (Fig. 3j). Collectively, these results demonstrate the potential of C-a16 LNPs in facilitating systemic mRNA delivery for both gene editing and protein supplementation therapies.

C-a16 LNP for tumour neoantigen vaccine delivery

Since we demonstrated that C-a16 LNPs can effectively traffic to lymph nodes after intramuscular injection (Supplementary Fig. 18), we evaluated if they could be used to deliver mRNAs encoding tumour antigens for cancer vaccination. We investigated the cell specificity of DiR-labelled C-a16 LNPs in lymph nodes after intramuscular injection (Supplementary Fig. 30). Our results indicated that C-a16 LNPs were mostly internalized by antigen-presenting cells, including dendritic cells (DCs), macrophages and B cells (Supplementary Figs. 30 and 31). In addition, the number of DiR-positive DCs, macrophages, B cells and T cells in the lymph nodes of C-a16 LNP-treated mice was approximately 1.73-fold, 1.56-fold, 3.56-fold and 1.81-fold higher, respectively, than that observed in mice treated with SM-102 LNPs that Moderna has used for the SARS-CoV-2 mRNA vaccines (Extended Data Fig. 2a–d). Furthermore, we tracked the biodistribution of DiR-labelled C-a16 LNPs in lymph nodes via fluorescence microscopy (Extended Data Fig. 2e). We found that DiR-labelled LNPs were internalized mainly by DCs. These results support the potential of C-a16 LNPs for effectively targeting DCs.

Next, we investigated whether C-a16 LNPs delivering mRNA encoding a model antigen could generate antigen-specific T cell responses. Mice were given intramuscular injections of C-a16 LNPs encapsulating mRNA encoding ovalbumin (OVA) at a dose of 0.25 mg kg^{-1} mRNA on day 1, with a booster shot administered on day 21. Seven days following the last vaccination, a flow cytometry experiment was performed to characterize antigen-specific T cells. C-a16 LNPs induced a significantly higher percentage of SIINFEKL (OVA MHC-I epitope)-Tetramer⁺ T cells (1.42%) in blood cells, whereas C-a16-Q LNPs and SM-102 LNPs resulted in lower ratios of SIINFEKL-Tetramer⁺ T cells, at about 0.3% and 0.6%, respectively (Extended Data Fig. 2f,g). Furthermore, we conducted an *in vivo* killing assay³⁶ to assess whether vaccination induced antigen-specific immune responses. We found that C-a16 LNP treatment led to approximately 70% killing of SIINFEKL⁺ target cells, whereas the benchmark SM-102 LNP and the control C-a16-Q LNP treatments only induced approximately 38% and 31% killing of target cells, respectively (Extended Data Fig. 2h,i). Since studies have shown that

longer durations of tumour antigen presence *in vivo* usually generate stronger antigen-specific immune responses, these results demonstrate that C-a16 LNPs, when used to deliver OVA mRNA, generated robust antigen-specific immune responses.

We then tested the antitumour efficacy of the C-a16 LNP (Fig. 4a–k). Using an OVA-expressing melanoma model (B16F10-OVA), we vaccinated mice with C-a16 LNPs encapsulating OVA mRNA (0.25 mg kg^{-1}) on days 7 and 14 post-tumour inoculation. The vaccination substantially inhibited tumour growth (Fig. 4b). While the tumours in the PBS group reached a size of $1,500 \text{ mm}^3$ (study endpoint) on day 28, the tumours in mice treated with C-a16 LNPs were only approximately 156 mm^3 . The tumour sizes were also smaller than those in mice treated with SM-102 LNPs and C-a16-Q LNPs (which showed tumour sizes of $\sim 526 \text{ mm}^3$ and $\sim 669 \text{ mm}^3$ for mice receiving SM-102 LNP and C-a16-Q LNP, respectively). In addition, C-a16 LNP treatment did not induce changes in body weight (Extended Data Fig. 3a) and greatly extended the survival time of mice when compared with those treated with PBS, SM-102 LNPs and C-a16-Q LNPs (Extended Data Fig. 3b).

While we have demonstrated great inhibition of B16F10-OVA tumour growth using C-a16 LNPs, it is important to note that the tumour model that we used overexpresses the OVA antigen, which is not representative of the tumours typically found in patients³⁷, as most do not exhibit such overexpression of antigens. Recently, neoantigen vaccines^{38,39} have shown great promise in clinical trials for cancer treatment^{40,41}. Therefore, we conducted experiments to assess whether C-a16 LNPs, delivering mRNA encoding a tumour neoantigen (Pbk-Actn4)⁴², could also inhibit tumour growth in a tumour model without OVA overexpression. The C-a16 LNP, delivering Pbk-Actn4 mRNA, greatly suppressed B16F10 tumour growth (Fig. 4c) and prolonged the survival of mice, as shown in Extended Data Fig. 3c, without causing a loss in mouse body weight (Fig. 4d). Furthermore, tumours in mice treated with C-a16 LNPs were notably smaller than those in mice treated with SM-102 LNPs and C-a16-Q LNPs at day 28 (Fig. 4c). We then explored whether the inhibition of tumour growth was a result of generating tumour-specific immune responses, as detailed in Fig. 4e–k and Supplementary Figs. 32 and 33. C-a16 LNP treatment significantly increased the infiltration of total T cells (Fig. 4e,f), especially of CD8⁺ T cells within the tumour tissue (Fig. 4g,h and Supplementary Fig. 33a). In addition, the levels of T cell exhaustion in the tumour tissue were reduced, characterized by lower expression of PD-1 on T cells (Fig. 4i and Supplementary Fig. 33c) and decreased levels of regulatory T cells (CD4⁺CD25⁺FoxP3⁺) (Fig. 4j and Supplementary Fig. 33d). Moreover, we observed a notable increase in effector memory (CD44^{high}CD62^{low}) and central memory (CD44^{high}CD62^{high}) cell populations in C-a16 LNP-treated mice (Fig. 4k and Supplementary Fig. 33e), which contributed to the effective rejection of tumour cells upon rechallenge (Supplementary Fig. 33f). These antitumour responses were further confirmed through hematoxylin and eosin staining and immunohistochemical staining (Supplementary Fig. 34). Altogether, these results suggest that C-a16 LNPs hold substantial promise as a technology for delivering neoantigen mRNA vaccines.

C-a16 LNP for SARS-CoV-2 vaccine delivery

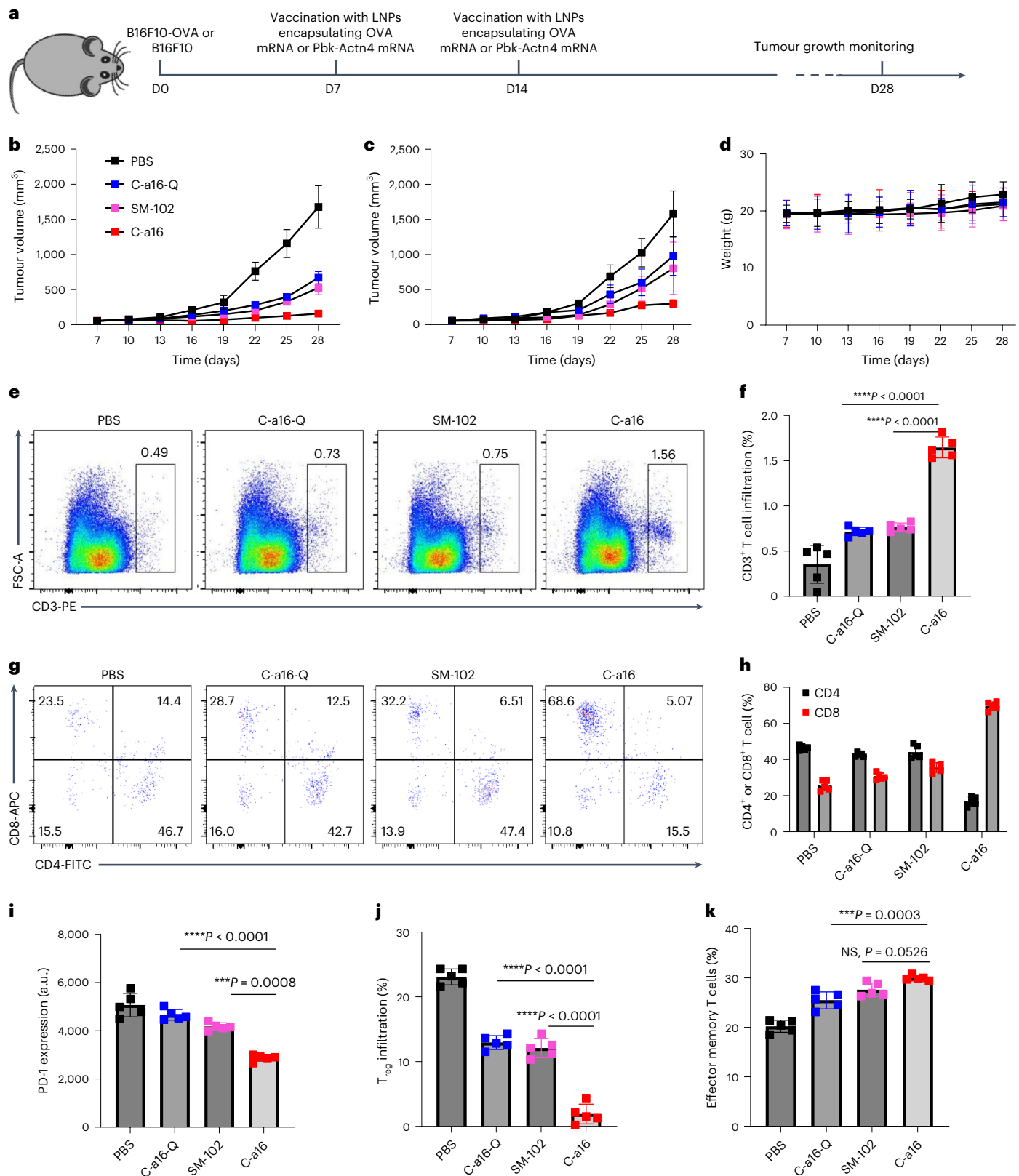
We further evaluated the efficacy of C-a16 LNPs as vaccines delivering mRNA encoding a SARS-CoV-2 spike protein and examined

Fig. 4 | Inhibition of tumour growth by C-a16 LNP encapsulating an OVA mRNA or a neoantigen mRNA. **a**, Mice were subcutaneously injected with either B16F10 cells expressing OVA or B16F10 cells on day 0. LNPs encapsulating mRNA encoding OVA or the neoantigen Pbk-Actn4 were administered intramuscularly on days 7 and 14 (0.25 mg kg^{-1}). **b–d**, Tumour growth (**b,c**) and changes in body weight were monitored (**d**). To investigate if tumour growth inhibition resulted from the activation of tumour-specific immune responses, a separate animal experiment was conducted. B16F10 tumour-bearing mice were treated as described above. On day 28, mice were euthanized. Tumour tissue was collected, digested into a single-cell suspension, stained with antibodies and analysed

by flow cytometry. **e**, Representative flow dot plots depicting the infiltration of CD3⁺ T cells in tumour tissues. **f**, Quantification of CD3⁺ T cell infiltration. **g**, Representative flow dot plots showing CD4⁺ and CD8⁺ T cell infiltration in tumour tissues. **h**, Quantification of CD8⁺ and CD4⁺ T cells in the tumour tissue. **i–k**, Quantification of PD-1 expression levels in tumour infiltrating T cells (**i**), regulatory T cell (T_{reg}) infiltration levels (**j**) and effector memory T cell infiltration levels (**k**) in tumour tissues. Data in **b–d** are presented as mean $\pm$ s.d. ($n = 8$ biological replicates). Data in **f** and **h–k** are presented as mean $\pm$ s.d. ($n = 5$ biological replicates). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.

the cellular and humoral immune responses. Mice received immunization on days 0 and 21. Throughout the treatment period, mice did not show body weight loss. Fourteen days after the second dose, we collected spleens from the mice and incubated splenocytes with a peptide pool containing SARS-CoV-2 receptor binding domain (RBD) peptides. Flow cytometry was used (Supplementary

Fig. 35) to determine the levels of antigen-specific CD4⁺ and CD8⁺ T cells. All LNPs elicited CD4⁺ T cells specific to RBD that expressed Th1 cytokines⁴³, including interferon (IFN)- γ , IL-2 and TNF- α (Fig. 5b–d). However, the responses induced by C-a16 LNPs were significantly more robust than those triggered by SM-102 LNPs and C-a16-Q LNPs (Fig. 5b–d). In addition, none of the vaccine



formulations induced CD4⁺ T cells expressing type 2 (Th2) cytokines (IL-4 and IL-5)⁴⁴ and IL-17 (Extended Data Fig. 4a–c), indicating a pre-dominant Th1-biased T cell immune response after vaccination. Furthermore, C-a16 LNPs also stimulated a greater number of RBD-specific CD8⁺ T cells that expressed Th1 immune response cytokines (IFN- γ , IL-2 and TNF- α ; Fig. 5e–g) and the cytotoxic marker CD107 α (Extended Data Fig. 4d). These results suggest that C-a16 LNPs can induce more potent RBD-specific Th1 and CD8⁺ T cell immune responses. Moreover, C-a16 LNPs led to a significantly higher proportion of polyfunctional^{45,46} CD4⁺ and CD8⁺ T cells^{36,37} compared with SM-102 LNPs (Extended Data Fig. 5). The pattern of cytokine secretion among these double-positive cells differed between CD4⁺ (IL-2⁺ TNF- α ⁺) and CD8⁺ (IFN- γ ⁺ TNF- α ⁺) cells and was consistent with the respective helper and cytotoxic functions of these two distinct lymphocyte populations.

Subsequently, we examined the humoral immune responses triggered by different LNP vaccine formulations (Supplementary Fig. 36). We conducted an endpoint dilution ELISA on serum samples from vaccinated mice to determine the total IgG antibody titres specific to the RBD. All three LNP vaccines induced substantial levels of RBD-specific IgG antibodies (Fig. 5h–j). Notably, the IgG titre for the C-a16 LNP vaccine (4.1×10^6) was approximately 5 times higher than that of the SM-102 LNP (8.8×10^5) and the C-a16-Q LNP (8.1×10^5) vaccines. In addition, ELISA analysis of RBD-specific IgG isotypes showed that the C-a16 LNP vaccine generated higher IgG1 and IgG2c titres when compared with the SM-102 LNP vaccine (Extended Data Fig. 6a). This further emphasizes that C-a16 LNPs, with their enhanced and durable antigen expression, have a tendency to elicit stronger RBD-specific immune responses. We then investigated whether the RBD-specific IgG antibodies induced by the LNP vaccines could effectively neutralize a vesicular stomatitis virus (VSV)-based pseudovirus (Fig. 5k). The C-a16 LNP vaccine produced markedly higher levels of neutralizing antibodies (NAbs) against SARS-CoV-2 pseudoviruses compared with the SM-102 LNP and C-a16-Q LNP vaccines, as demonstrated by the higher 50% foci reduction neutralization titres (FRNT50) (Fig. 5k).

We subsequently examined whether C-a16 LNPs could stimulate the generation of B cell memory. We found that C-a16 LNP vaccination significantly increased the number of RBD-specific B cells compared with SM-102 LNPs (Extended Data Fig. 6b), and these cells showed a memory phenotype (CD38⁺GL7⁺; Extended Data Fig. 6c). Three months after the second vaccination, we conducted an analysis of long-lived plasma cells (LLPCs)⁴⁷ residing in the bone marrow. These LLPCs are a cell type responsible for providing lasting protection against infection by continually producing antibodies specific to the antigen. We collected bone marrow samples from immunized mice and used an enzyme-linked immunospot (ELISpot) assay to investigate different subsets of antibody-secreting cells (ASCs) targeting the RBD. As shown in Fig. 5l–n, while the treatment with SM-102 LNPs led to a modest increase in the generation of RBD-specific IgG1-, IgG2b- and IgG2c-expressing ASCs (Fig. 5l–n), C-a16 LNPs resulted in a 1.8-fold, 1.9-fold and 1.7-fold increase in IgG1-, IgG2b- and IgG2c-expressing ASCs, respectively. Moreover, we also found that C-a16 LNP treatment did not lead to changes in tissue structures (Supplementary Fig. 37). Altogether, these results demonstrate that the C-a16 vaccine triggers more robust humoral immune and LLPC responses than SM-102.

Outlook

This study shows the utility of Mannich reactions in the construction of combinatorial libraries for the synthesis of ionizable lipids. Through in vitro and in vivo high-throughput screening, we identified an antioxidant ionizable lipid C-a16 for mRNA delivery with low immunogenicity. C-a16 LNPs effectively mitigate intracellular ROS, reducing cellular inflammatory responses and extending the duration of intracellular protein expression. To assess its potential, we compared C-a16 LNPs with DLin-MC3-DMA LNPs (the formulation used in Alnylam's intravenously administered siRNA drug Onpattro) for systemic delivery applications and found about 2.8-times higher *Ttr* gene editing. We also compared C-a16 LNPs with SM-102 LNPs (the formulation used in Moderna's intramuscularly administered COVID-19 vaccine) for tumour vaccine and SARS-CoV-2 vaccine applications and found substantial improvements in the inhibition of tumour growth and the generation of RBD-specific cellular and humoral immune responses, respectively. Future studies will explore the application of this C-a16 LNP technology in various applications, including flu vaccines and autoimmune disease⁴⁸ vaccines, as well as base editor⁴⁹ delivery. We believe that this study will inspire further exploration at the intersection of cellular defence and nucleic acid delivery.

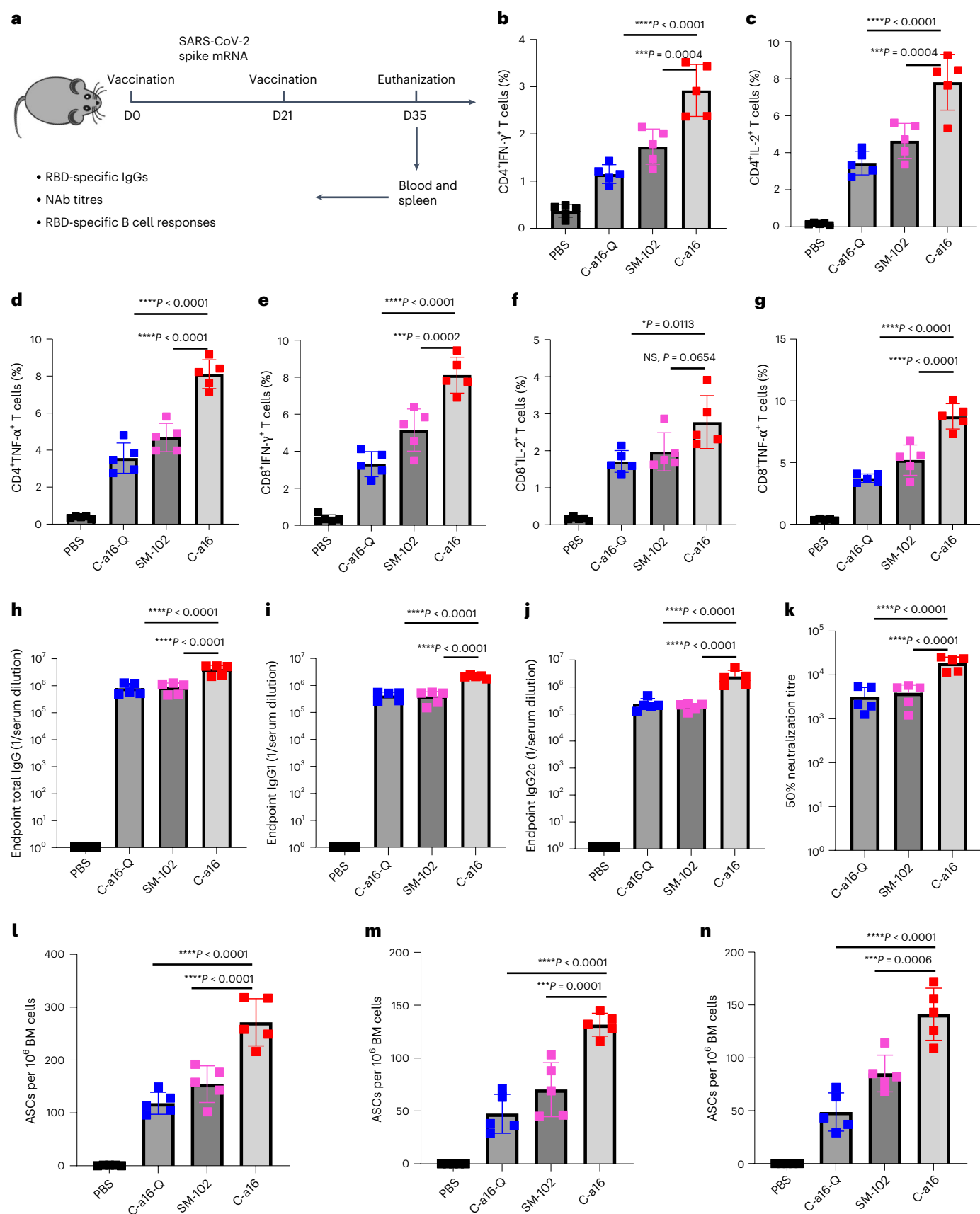
Methods

Chemicals and antibodies

SM-102 lipid was purchased from Cayman Chemical. DLin-MC3-DMA was obtained from MedChem Express. 1,2-Distearoyl-sn-glycero-3-phosphocholine (DSPC), DOPE, cholesterol and DMG-PEG 2000 were purchased from Avanti Polar Lipids. Chemicals for the synthesis of various ionizable lipids were purchased from Sigma-Aldrich, TCI Chemicals, Ambeed or ThermoFisher. Antibodies including rat anti-mouse CD107a-FITC (clone 1D4B, catalogue number 553793, 1:100 dilution), rat anti-mouse IL-4-BV786 (clone 11B11, catalogue number 564006, 1:100 dilution), rat anti-mouse IL-2-APC (clone JES6-5H4, catalogue number 554429, 1:100 dilution), rat anti-mouse IFN- γ -AF700 (clone XMG1.2, catalogue number 557998, 1:100 dilution), rat anti-mouse TNF-PE/Cy7 (clone MP6-XT22, catalogue number 557644, 1:100 dilution), hamster anti-mouse CD80-BV650 (clone 16-10A1, catalogue number 563687, 1:100 dilution), rat anti-mouse IgM-PE/CF594 (clone R6-60.2, catalogue number 562565, 1:100 dilution) and rat anti-mouse CD8a-PE/Cy5 (clone 53-6.7, catalogue number 553034, 1:100 dilution) were purchased from BD Biosciences. Anti-mouse CD4-PerCP/Cy5.5 (clone GK1.5, catalogue number 100434, 1:100 dilution), anti-mouse CD8a-Pacific Blue (clone 53-6.7, catalogue number 100725, 1:100 dilution), anti-mouse CD3e-BV605 (clone 145-2C11, catalogue number 100351, 1:100 dilution), Zombie Aqua fixable viability kit (catalogue number 423102, 1:1,000 dilution), anti-mouse IgD-BV711 (clone 11-26 c.2a, catalogue number 405731, 1:100 dilution), anti-mouse/human CD45R/B220-BV785 (clone RA3-6B2, catalogue number 103246, 1:100 dilution), anti-mouse CD273-PE (clone TY25, catalogue number 107206, 1:100 dilution), anti-mouse CD4-PE/Cy5 (clone H129.19, catalogue number 130312, 1:100 dilution), anti-mouse Ly-6G/Ly-6C (Gr-1)-PE/Cy5 (clone RB6-8C5, catalogue number 108410, 1:100 dilution), anti-mouse/human GL7-PE/Cy7 (clone GL7, catalogue number 144620, 1:100 dilution), BV421 streptavidin (catalogue number 405225, 1:5,000 dilution) and AF647 streptavidin (catalogue number 405237, 1:5,000 dilution) were purchased from BioLegend. Live/Dead

Fig. 5 | Immune responses elicited by C-a16 LNP encapsulating mRNA encoding SARS-CoV-2 spike protein. **a**, Illustration of the prime and boost vaccination strategy and immune response analysis. Mice received intramuscular immunization with various LNP vaccines on days 0 and 21 (0.25 mg kg⁻¹ mRNA). On day 35, splenocytes were collected and stimulated with SARS-CoV-2 RBD peptide pools. The expression of Th1 intracellular cytokines (IFN- γ , IL-2 and TNF- α) in T cells was assessed. **b–g**, Flow cytometry analysis of polyfunctional CD4⁺ (CD4⁺IFN- γ ⁺ (**b**), CD4⁺IL-2⁺ (**c**), CD4⁺TNF- α ⁺ (**d**)) and CD8⁺ (CD8⁺IFN- γ ⁺ (**e**),

CD8⁺IL-2⁺ (**f**), CD8⁺TNF- α ⁺ (**g**)) T cells. Humoral immune responses induced by the C-a16 LNP vaccine were also assessed. **h–j**, Serum collected from vaccinated mice on day 35 was used to determine RBD-specific total IgG (**h**), IgG1 (**i**) and IgG2c (**j**) levels by endpoint dilution ELISA. **k**, Measurement of neutralizing antibody levels using a pseudovirus neutralization assay. **l–n**, Quantification on day 120 of RBD-specific IgG1 (**l**), IgG2b (**m**) and IgG2c (**n**) ASCs in bone marrow by an ELISpot assay. Data in **b–n** are presented as mean $\pm$ s.d. ($n = 5$ biological replicates). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.



fixable aqua dead cell stain kit (catalogue number L34966, 1:1,000 dilution), anti-mouse CD38-AF700 (catalogue number 56-0381-82, 1:100 dilution), anti-mouse F4/80-PE-Cy5 (clone BM8, catalogue number 15-4801-82, 1:100 dilution) and anti-mouse CD19-PE/Cy5.5 (clone 1D3 catalogue number 35-0193-82, 1:100 dilution) were purchased from ThermoFisher. Biotinylated SARS-CoV-2 (2019-nCoV) Spike RBD-His Recombinant Protein (catalogue number 40592-V08H-B) was purchased from SinoBiological.

Cell lines and animals

The MC38, B16F10 and B16F10-OVA cell lines were originally obtained from ATCC. All cells were tested mycoplasma negative before use.

The 6–8-week-old female C57BL/6J mice and Ai9 mice (around 20 g) were purchased from Jackson laboratory. All mice were housed in a specific-pathogen-free animal facility at ambient temperature ($22 \pm 2^\circ\text{C}$), air humidity 40–70% and 12 h dark/12 h light cycle. All protocols performed on animals in this study were approved by the Institutional Animal Care and Use Committee of the University of Pennsylvania.

Synthesis of lipids via Mannich reaction

For the general Mannich reaction, phenols, thiols, alkyl ketones, pyroles and acetophenones derivatives were mixed with formaldehyde and amines at specific molar ratios in either ethanol or a mixture of ethanol and water. The reaction mixtures were heated at 80°C for 1–24 h. Solvents were evaporated under reduced pressure to obtain the crude product.

Example synthesis of C-a16. For large-scale synthesis of C-a16, *p*-octylphenol (4.5 mmol) was dissolved in 2 ml ethanol and the mixture was added into a 10 ml round-bottom flask. After that, a-16 (1.0 mmol) was added and the mixture was stirred at room temperature for 10 min. Formaldehyde solution (4.2 mmol) was added dropwise with magnetic stirring. The reaction mixture was then heated to 80°C and maintained for 24 h with stirring. After that, solvent was evaporated and crude compound was obtained. Purified compound was obtained by column chromatography using a mixture of isopropanol and acetonitrile (9:1) as the eluent.

Example synthesis of C-a16-Q (C-a16 with quenched phenol group). C-a16 (1.0 mmol) was dissolved in 5 ml dichloromethane and the mixture was added into a 10 ml round-bottom flask. Acetyl chloride (4.2 mmol) was then added dropwise when maintaining the reaction at 0°C (on ice). After that, the mixture was stirred for 24 h at room temperature. The mixture was washed with saturated sodium bicarbonate solution (3 times, 5 ml for each time) to neutralize any residue acetyl chloride, followed by washing with water (3 times, 5 ml for each time) to remove sodium bicarbonate. The organic layer was combined and dried with anhydrous sodium sulfate. After that, solvent was removed under reduced pressure and the crude product was obtained. Pure C-a16-Q was obtained by column chromatography using a mixture of isopropanol and acetonitrile (8:2) as the eluent. ^1H nuclear magnetic resonance (NMR) spectrum of compounds was collected using a Bruker 400 MHz NMR spectrometer equipped with a TopSpin software. Data were analysed using a MestReNova 9.0 software.

Synthesis of mRNAs

Codon-optimized firefly luciferase sequence, Cre recombinase sequence, human FGF21 sequence, tumour neoantigen Pbk-Actn4 sequence or SARS-CoV-2 spike protein sequence was cloned into an mRNA production plasmid (optimized 3' and 5' UTR and containing a 101 polyA tail). The plasmid was transcribed in vitro in the presence of 1-methyl pseudouridine-modified nucleoside, co-transcriptionally capped using the CleanCap technology (TriLink) and purified with cellulose to remove double-stranded RNAs. Purified mRNAs were

precipitated in ethanol, washed with ethanol and were re-suspended in nuclease-free water. Electrophoresis, dot blot and endotoxin assays were performed for quality control. All mRNAs were stored in a -80°C freezer until use.

Preparation of LNPs

For the preparation of DLin-MC3-DMA LNPs, DLin-MC3-DMA, DSPC, cholesterol and DMG-PEG were mixed at a molar ratio of 50/10/38.5/1.5 and dissolved in 200 proof ethanol. For the preparation of other LNPs, ionizable lipid, DOPE, cholesterol and DMG-PEG were mixed at a molar ratio of 50/10/38.5/1.5 and dissolved in ethanol. mRNA was dissolved in 10 mM citrate buffer (pH = 3). For in vitro screening experiments, the ethanol phase and the citrate buffer phase, containing lipid and mRNA at a mass ratio of 20:1, were mixed using pipette mixing. For other in vitro experiments and animal experiments, the ethanol phase and the citrate buffer phase were mixed in a Micro&Nano microfluidic chip device (iNano L⁺) at a flow rate ratio of 1:3. LNPs were then purified by dialysing against $1\times$ PBS (dialysis cassette molecular weight cut-off 20 kDa) for 2 h to remove ethanol and citrate buffer. A 0.22 μm pore size filtration device was used to remove any aggregates, and the as-obtained LNPs were stored in a 4°C fridge until use. A Quant-iT RiboGreen RNA assay was used to quantify the mRNA encapsulation efficiency. mRNA encoding firefly luciferase was used to formulate the LNPs for in vivo screening experiments.

Labelling LNPs with DiR/DiO

One millilitre of LNPs was added to a 1.5 ml tube with 2 μl DMSO containing DiR/DiO (1 mg ml^{-1}). After vortex mixing and incubation at room temperature (25 min), the mixture was loaded on a centrifugal filter device (MWCO 10 kDa) and LNP-DiR/DiO was washed with PBS 3 times to remove free DiR/DiO. The purified LNP-DiR nanoparticles were stored in a 4°C fridge before use.

Characterization of the nanoparticles

The size and PDI of LNPs were determined using a Malvern Nano ZS90. The cryo-EM images of LNPs were collected using a Krios cryo-EM (ThermoFisher).

IVIS imaging

A PerkinElmer Lumina III IVIS was used to collect the IVIS images shown in this study. Images were collected under the defined luminescence channel using the auto exposure mode. LivingImage 4.5 software and the Auto region of interest function were used to quantify the luminescence level in each image.

RNA sequencing

MC38 cancer cells were seeded in a 6-well plate (5×10^6 cells per well) and were treated with PBS, DLin-MC3-DMA LNPs or C-a16 LNPs for 5 h. After that, the cells were washed with PBS three times and total RNA was extracted. Sequencing was performed in Novogene and the differentially expressed genes (DEGs) were analysed using DEGSeq R package (1.20.0). The *P* values were adjusted using the Benjamini and Hochberg method. The threshold of corrected *P* value of 0.05 and $\log_2$ (fold change) of 1 was selected to determine the significant DEGs.

Flow cytometry

Mice were perfused with 50 ml PBS from the left ventricle to remove blood. Liver, spleen or tumour tissues were isolated and were cut into $3 \times 3 \times 3\text{ mm}$ cubes. After enzymatic digestion with collagenase V (1 mg ml^{-1}), RNase (40 U ml^{-1}) and DNase I (40 U ml^{-1}) for 30 min at 37°C , tissues were ground on a 70 μm cell strainer to obtain single-cell suspensions. The cells were washed three times and were stained with Live/Dead fixable aqua dead cell stain kit (ThermoFisher, L34966). Cellular surface staining was performed for 30 min on ice. After that, cells were fixed with 4% PFA and then washed 3 times with PBS. After that, cells were stained

with antibodies for intracellular staining, washed with PBS and analysed using flow cytometry (BD LSR II) equipped with a Diva 6.0 software. Flow cytometry data were analysed using a FlowJo V10 software.

Immunofluorescence sample preparation

A transcardiac perfusion was performed by slow injection of 50 ml 10% neutral buffered formalin into the left ventricles of mice. After that, organs were collected. Frozen sections were put onto slides. The slides were stained with different primary antibodies and dye-conjugated secondary antibodies before observing under confocal microscopy equipped with a ZEN2010 software.

In vivo vaccination

Mice received intramuscular immunization with SARS-CoV-2 mRNA-loaded LNPs at 0.25 mg kg⁻¹ mRNA dose, following a prime-boost strategy with a 3-week interval between doses. Mouse body weight was documented twice weekly throughout the course of the experiment. Serum samples were obtained using serum separator tubes, as detailed previously, and stored at -20 °C for subsequent use in ELISA and virus neutralization assays. Two weeks after the booster vaccination, the mice were euthanized, and spleens were collected for analysis via flow cytometry.

Pseudovirus neutralization assay

A VSV pseudotype of the SARS-CoV-2 S protein (provided by ReVaccSci) was used for conducting a neutralization antibody assay. A 96-well plate was used to culture A549 cells that stably expressed the human ACE2 receptor (A549-ACE2). After a 12 h incubation, serum samples were diluted and combined with the VSV pseudotype of the SARS-CoV-2 S virus (diluted 20 times) and then incubated at 37 °C for 1 h. To prevent any potential carryover of VSV-G, a mouse anti-VSV Indiana G antibody, 8G5F11 (#Ab01401-2.0, supplied by Absolute Antibody), was added to the mixture. Subsequently, this antibody-virus mixture was applied to treat A549-ACE2 cells. After 24 h of infection, the culture medium was removed, and the expression of luciferase in the cells was quantified using a Luciferase 1000 assay. The focus reduction neutralization titre 50% (FRNT50) was determined as the highest serum dilution at which luciferase expression decreased by at least 50% compared with control cells that were infected with the pseudotype virus in the absence of mouse serum.

Flow cytometry analysis of vaccine potency

To evaluate T cell activations, spleens were collected, processed into single-cell suspensions and treated with ACK lysis buffer to remove red blood cells. To assess antigen-specific T cells, 2 × 10⁶ splenocytes were activated with SARS-CoV-2 RBD peptide pools (#PM-WCPV-S-RBD-1, JPT) at a concentration of 2.5 µg ml⁻¹, while being simultaneously exposed to 2 mg ml⁻¹ of anti-CD28 antibody for co-stimulation. Following a 1 h stimulation period, 5 mg ml⁻¹ of brefeldin A (BioLegend #420601) and 2 mM of monensin (BioLegend #420701) were introduced, and stimulation continued for an additional 5 h. Subsequently, the cells were rinsed with PBS, stained with Live/Dead dye, blocked with anti-mouse CD16/32 antibody and stained with various antibodies (30 min). The cells were then rinsed, fixed, permeabilized and internally stained with different antibodies for 30 min. After intracellular staining, the cells were washed twice with PBS and fixed with 1% PFA. Flow cytometry data were acquired using a BD LSR II flow cytometer.

To assess memory B cells, 2 million splenocytes were blocked with CD16/32 antibody and stained with various antibodies for 30 min. Subsequently, the cells were rinsed with PBS and fixed with 1% PFA. Flow cytometry data were collected using a BD LSR II flow cytometer.

ELISpot assay

Bone marrow was extracted from the femurs and tibia of mice, followed by filtration through a 70 µm cell strainer. To remove red blood

cells, the bone marrow was exposed to ACK buffer for 5 min at room temperature, followed by two PBS washes. Plates were prepared by coating them with the RBD protein at a concentration of 10 µg ml⁻¹ in a sodium carbonate/sodium bicarbonate buffer (pH 9.6) for 1 h at 37 °C. Subsequently, the plates underwent 3 PBS washes and were blocked at 37 °C using complete RPMI for 30 min. Bone marrow cells were plated in six halving dilutions, starting with one million total bone marrow cells per well. After an overnight incubation, the plates were washed, and a biotinylated anti-IgG detection antibody (goat anti-mouse IgG human ads-BIOT; Southern Biotech, 1030-08) was applied at a final dilution of 3 µg ml⁻¹, followed by a 1 h incubation at room temperature. The plates were washed five times, and streptavidin-alkaline phosphatase (diluted 1:20,000 in 2% BSA/PBS) was added before a 30 min incubation at room temperature. Following another five washes with a wash buffer, a solution of 5-bromo-4-chloro-3-indolyl-phosphate/nitro blue tetrazolium chloride (50 µl per well, Sigma, #B1911) was added until spots developed, after which the reaction was halted with 100 µl of a 1 M sodium phosphate monobasic solution. The plates were then rinsed with deionized water, left to dry overnight, and subsequently scanned and counted.

Statistics

Statistical analyses were performed using GraphPad Prism 7.0 software. The error bars indicate the mean ± s.d. To assess statistical differences between two groups, Student's *t*-test was used. For evaluating statistical differences among multiple groups, a one-way ANOVA with Tukey's post hoc test was utilized. In animal survival experiments, the *P* values were determined using the log-rank test based on the Kaplan–Meier method.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the paper and its Supplementary Information files. Source data are provided with this paper.

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Acknowledgements

M.J.M. acknowledges support from a US National Institutes of Health (NIH) Director's New Innovator Award (DP2 TR002776), a Burroughs Wellcome Fund Career Award at the Scientific Interface (CASI), a US National Science Foundation CAREER Award (CBET-2145491) and the American Cancer Society (number RSG-22-122-01-ET). E.L.H. acknowledges support from a US National Science Foundation Graduate Research Fellowship (DGE 1845298). C.G.F.-E. acknowledges support from a US National Science Foundation Graduate Research Fellowship (DGE 1845298), a GEM Fellowship and the NIH/National Cancer Institute Pre-doc to Post-doc Transition Award (F99 CA284294).

Author contributions

N.G., D.K. and M.J.M. conceived of and designed the experiments. N.G., D.K., M.-G.A., R.E.-M., G.D., Q.S., X.H., L.X., J.X., R.P., Z.M., T.L. and C.G.F.-E. performed the experiments. N.G., D.K. and R.E.-M. analysed the data. D.W. and J.L. involved in results discussion. N.G., D.K., E.L.H.,

R.E.-M. and M.J.M. wrote and edited the paper. M.J.M. supervised the entire project. All authors discussed the results and commented on the paper.

Competing interests

N.G. and M.J.M. have filed a patent application related to this study. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41551-025-01422-8>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-025-01422-8>.

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Peer review information *Nature Biomedical Engineering* thanks Bruno De Geest, Shuai Liu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

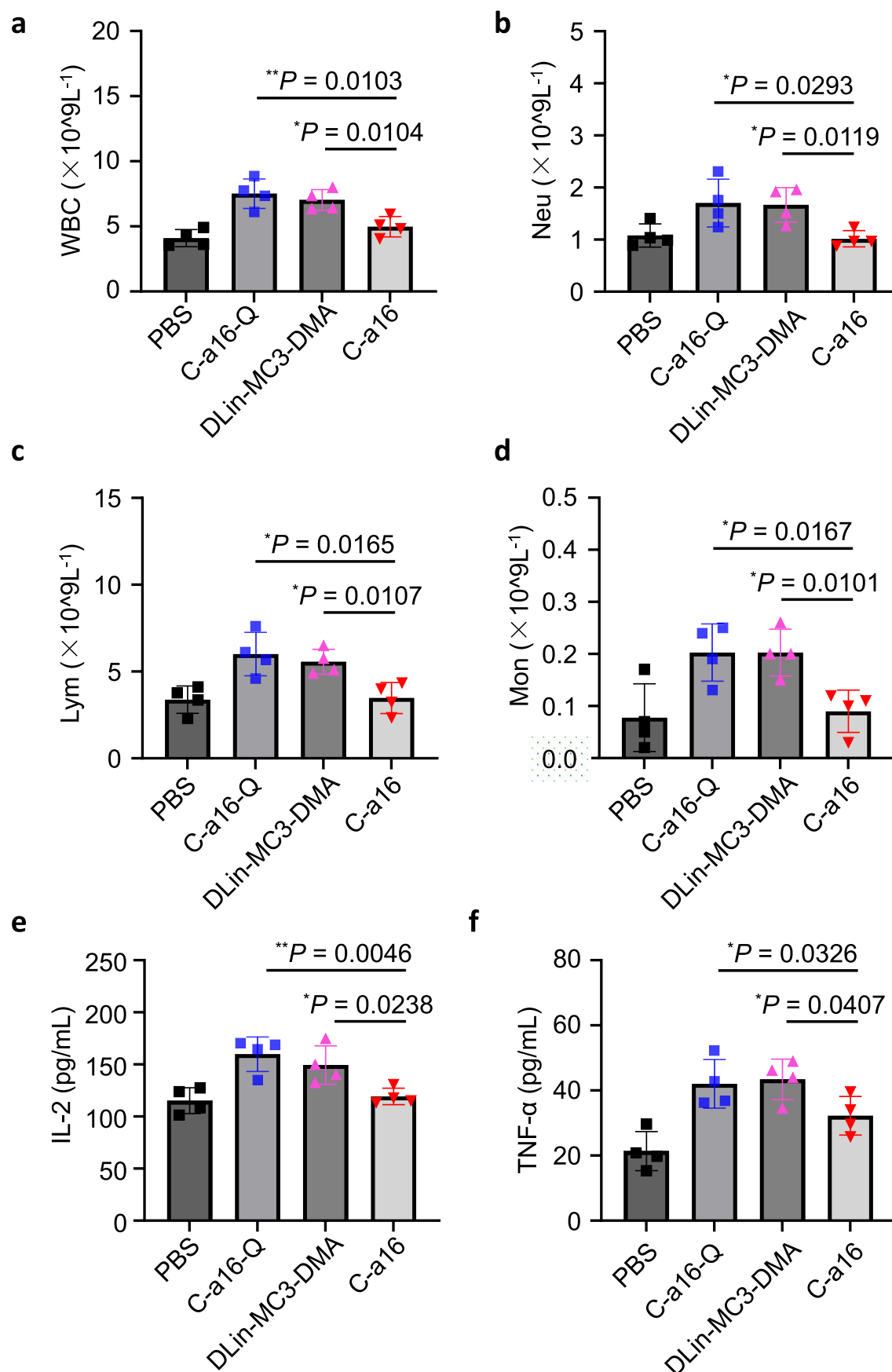
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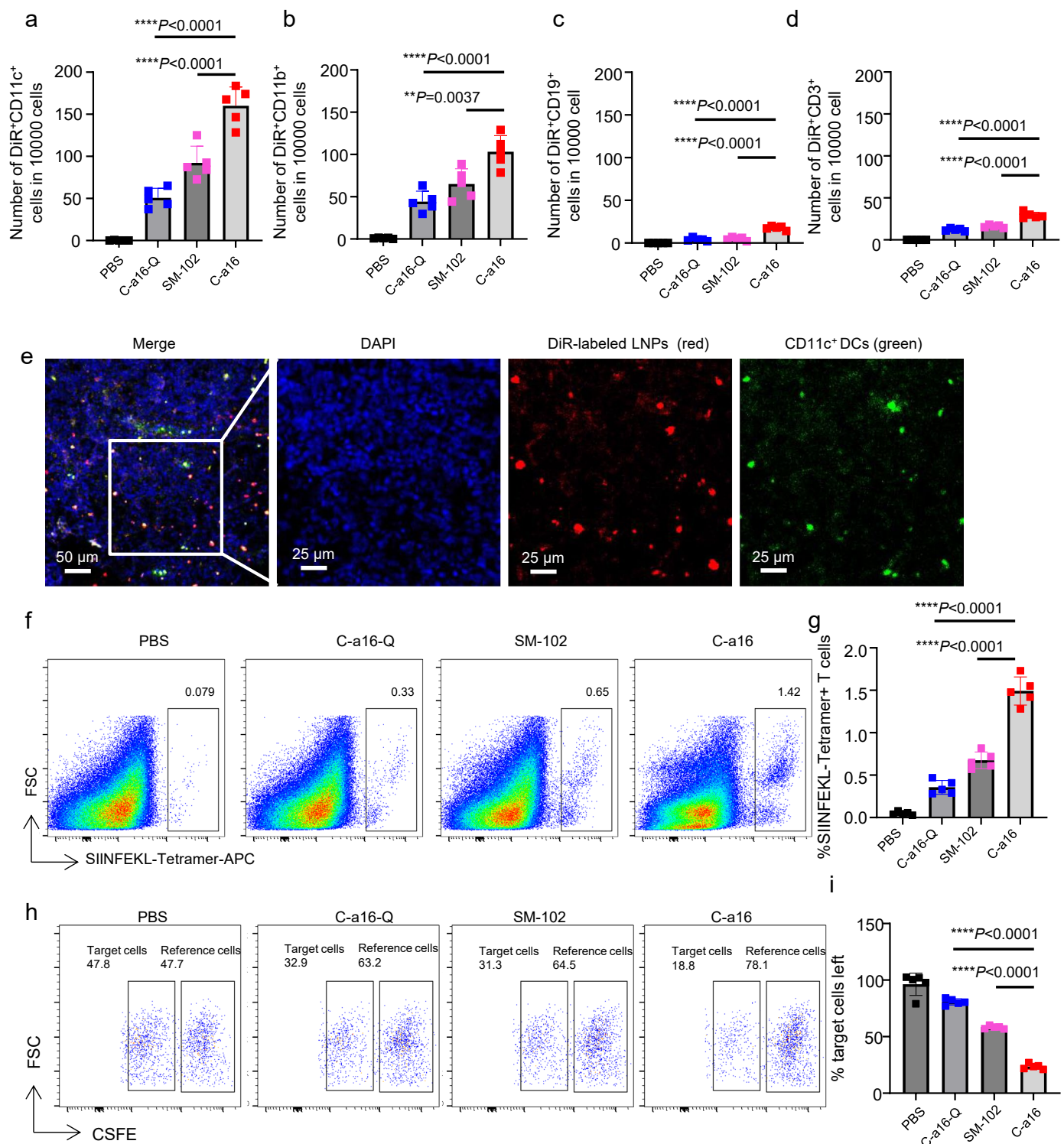
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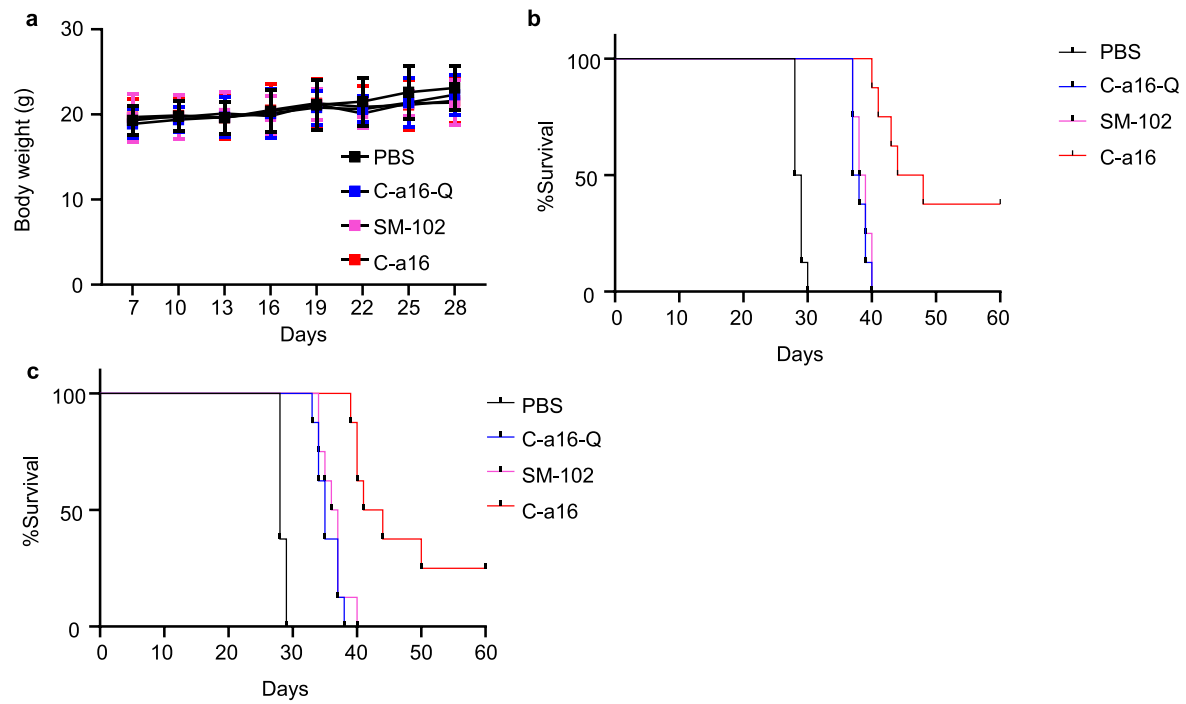
Extended Data Fig. 1 | Evaluation of LNP-induced innate immunity activation *in vivo* using complete blood counting assay and ELISA assay. C-a16, C-a16-Q, DLin-MC3-DMA LNPs or PBS was intravenously injected, and a complete blood counting (CBC) assay was performed 24 hours post-LNP injection. **a**, The levels of

white blood cells (WBC), **b**, neutrophils (Neu), **c**, lymphocytes (Lym), and **d**, monocytes (Mon), in the blood were measured ($n = 4$). ELISA assay was used to quantify the levels of IL-2 (**e**) and TNF- α (**f**) in the mouse serum ($n = 4$). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.

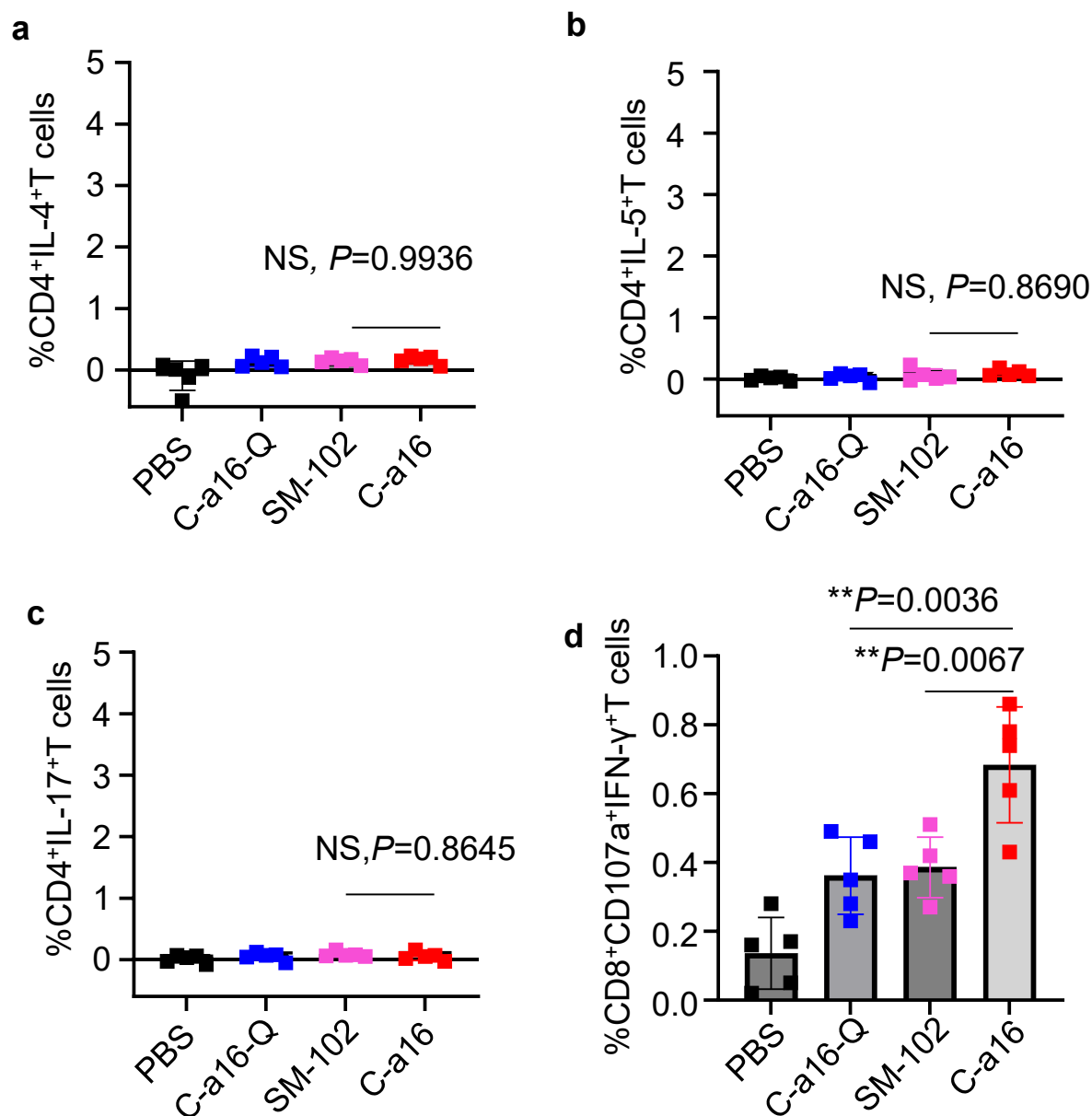
**Extended Data Fig. 2 | C-a16 LNPs induce antigen-specific immune responses.**

a–d, Quantification of LNP distribution in immune cell populations, including DCs (**a**), macrophages (**b**), B cells (**c**), and T cells (**d**), assessed by flow cytometry. Data are presented as mean $\pm$ SD ($n = 5$), and statistical differences were determined by one-way ANOVA. **e**, Immunofluorescence images of lymph nodes in mice after intramuscular injection of DiR dye-labeled LNPs. Nuclei are represented in blue, CD11c⁺ DCs in green, and DiR-labeled LNPs in red. **f**, Evaluation of OVA-specific T cell generation following two doses of C-a16-mOVA vaccines. Seven days after the boost dose, blood T cells were stained with

SIINFEKL-H2Kb tetramer-APC antibody. After that, SIINFEKL-H2Kb tetramer⁺ cells were detected by flow cytometry (**f**) and quantified (**g**). **h** and **i**, Investigation of vaccine-induced immune responses' ability to eliminate SIINFEKL epitope⁺ target cells. A mixture of CFSE^{low} SIINFEKL-loaded and CFSE^{high} non-loaded splenocytes (in a 1:1 ratio) was intravenously injected into vaccinated mice. After 24 hours, splenocytes were collected and analyzed by flow cytometry (**h**), and target cell killing was quantified (**i**). Data in **a–d**, **g** and **i** are shown as mean $\pm$ SD ($n = 5$). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.

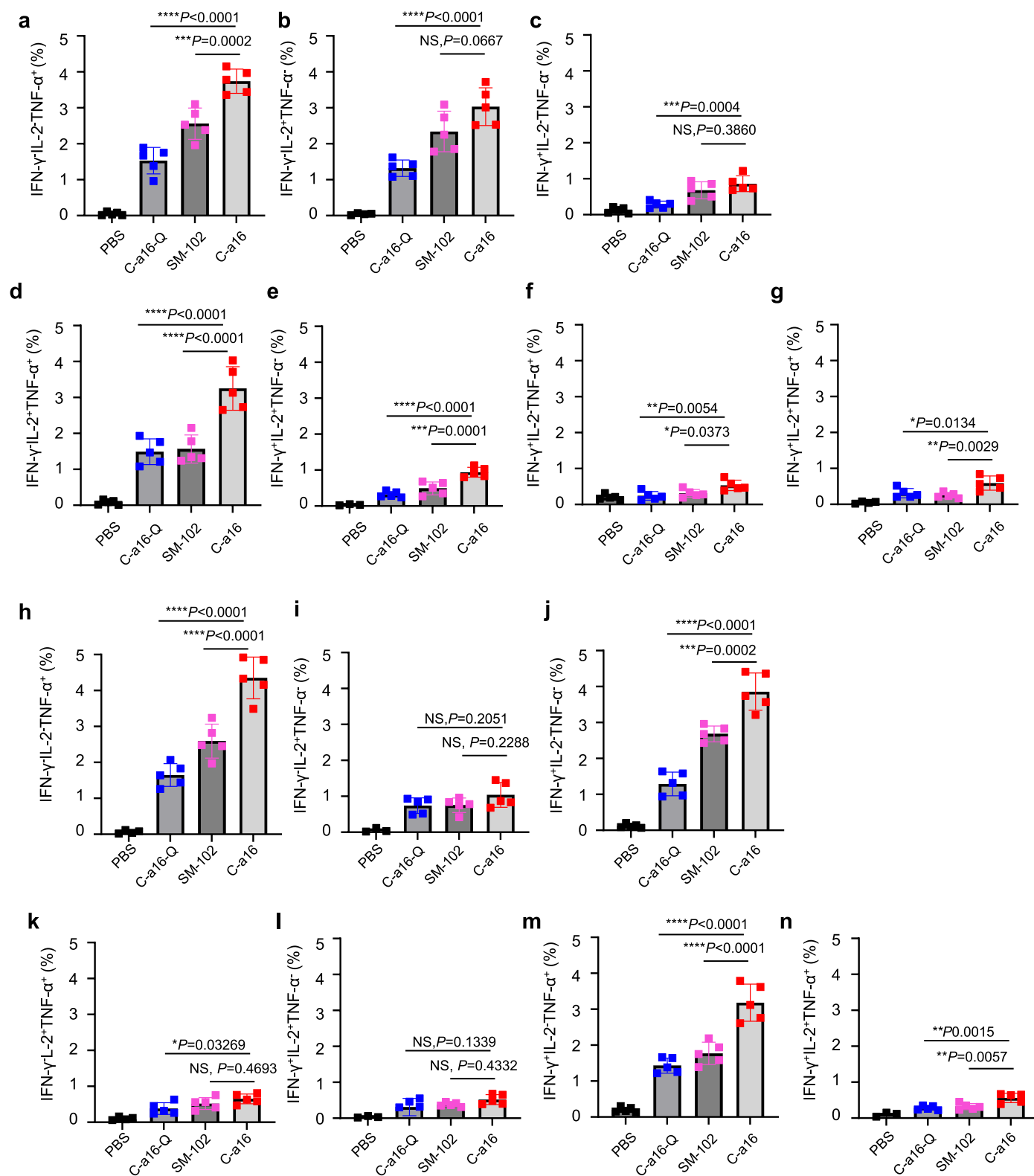


Extended Data Fig. 3 | Body weight change and survival curves. **a**, Body weight change after mice were treated with various LNP vaccines. **b**, Mice survival curves after treatment with various LNP vaccines encapsulating OVA mRNA. **c**, Survival curves of mice receiving various LNPs encapsulating Pbk-Actn4 mRNA. Data in **a** are shown as mean $\pm$ SD ($n = 8$).



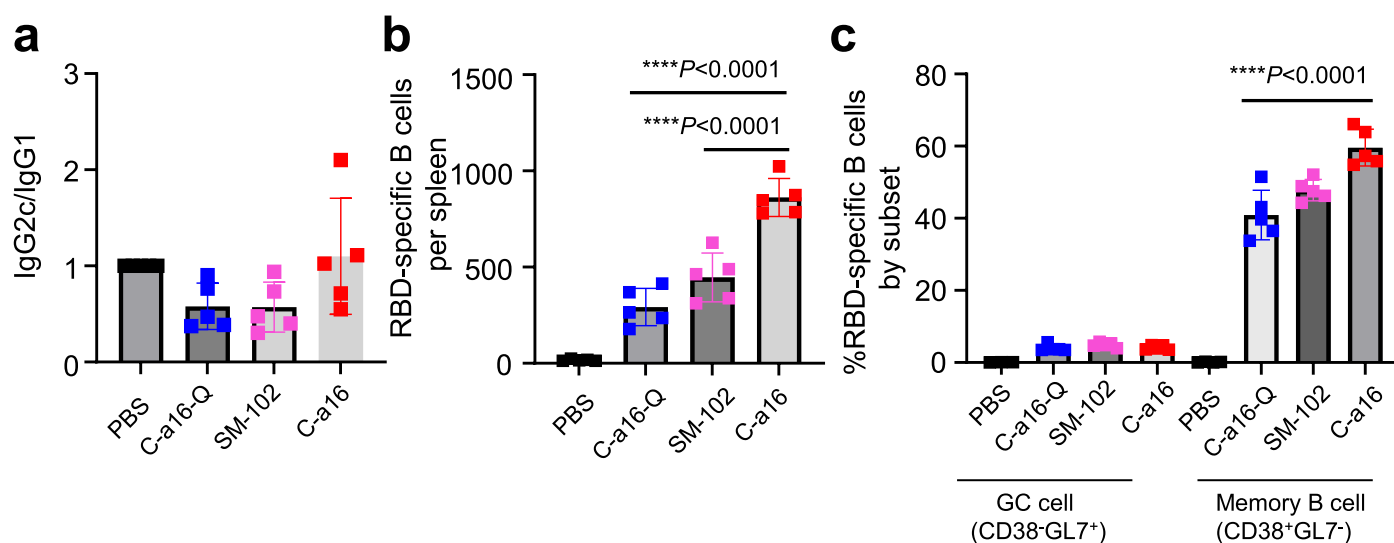
Extended Data Fig. 4 | T cell immune responses elicited by C-a16 LNP encapsulating mRNA encoding SARS-CoV-2 spike protein. Mice received intramuscular immunization with various LNP vaccines on days 0 and 21 (0.25 mg/kg). On day 35, splenocytes were collected and stimulated with SARS-CoV-2 RBD peptide pools. **a** and **b**, T cells were assessed for Th2 (IL-4, IL-5) and

c, Th17 (IL-17a) intracellular cytokine expression. **d**, Evaluation of the expression of the cytotoxic marker CD107a expression in CD8⁺ T cells. Data are shown as mean $\pm$ SD ($n=5$). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.

**Extended Data Fig. 5 | Evaluation of polyfunctional CD4⁺ and CD8⁺ T cells.**

Mice received intramuscular immunization with various LNP vaccines on days 0 and 21 (0.25 mg/kg). On day 35, splenocytes were collected and stimulated with

SARS-CoV-2 RBD peptide pools. CD4⁺ (a–g) and CD8⁺ (h–n) polyfunctional T cell percentages (%) were assessed. Data are shown as mean $\pm$ SD (n = 5). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.



Extended Data Fig. 6 | Investigation of humoral immune responses and memory B cells elicited by various LNPs. a, Mice received intramuscular immunization with various LNP vaccines on days 0 and 21 (0.25 mg/kg). IgG2c to IgG1 ratio was determined. Data are presented as mean $\pm$ s.d. **b,** Enumeration of RBD-specific B cells per spleen. Splenocytes were stimulated with SARS-CoV-2

RBD peptide pools and stained with various antibodies before flow cytometry analysis. **c,** Percentage of RBD-specific B cells categorized by germinal center (GC) or memory phenotype. GC B cells were defined as CD38⁺GL7⁺, and memory B cells were defined as CD38⁺GL7⁻. Data are shown as mean $\pm$ SD (n = 5). Statistical differences were calculated using one-way ANOVA with Tukey's post hoc test.

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Data analysis	Statistical analysis were performed on Graphpad Prism 7.0. Flow cytometry data were analyzed on FlowJo software package (FlowJo V10). ¹ H nuclear magnetic resonance (NMR) spectrum of compounds was analyzed using a MestReNova 9.0 software. Confocal data was analyzed using a ZEN2010 software.

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Sample size No effect size was predetermined, but sample sizes employed in this study are consistent with previously published works (Li A W, Sobral M C, Badrinath S, et al. Nature materials, 2018, 17(6): 528-534; or Kuai R, Ochyl L J, Bahjat K S, et al. Nature materials, 2017, 16(4): 489-496.). For example, in vitro studies were repeated at least three times independently and in the in vivo experiments with 7-10 mice per group were performed.

Data exclusions No animals and/or data were excluded.

Replication All experiments were repeated for at least three times and experimental findings were reproducible.

Randomization Samples were allocated into experimental groups randomly across all the in vitro and in vivo experiments in this study.

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☒ ☐ Clinical data

☒ ☐ Dual use research of concern

☒ ☐ Plants

Methods

n/a Involved in the study

☒ ☐ ChIP-seq

☐ ☒ Flow cytometry

☒ ☐ MRI-based neuroimaging

Antibodies

Antibodies used Rat anti-Mouse CD107a-FITC (clone: 1D4B, catalog number: 553793, 1:100 dilution), rat anti-mouse IL-4-BV786 (clone: 11B11,

Antibodies used

catalog number: 564006, 1:100 dilution), rat anti-mouse IL-2-APC (clone: JES6-5H4, catalog number: 554429, 1:100 dilution), rat anti-mouse IFN- γ -AF700 (clone: XMG1.2, catalog number: 557998, 1:100 dilution), rat anti-mouse TNF-PE/Cy7 (clone: MP6-XT22, catalog number: 557644, 1:100 dilution), hamster anti-mouse CD80-BV650 (clone: 16-10A1, catalog number: 563687, 1:100 dilution), rat anti-mouse IgM-PE/CF594 (clone: R6-60.2, catalog number: 562565, 1:100 dilution), and rat anti-mouse CD8a-PE/Cy5 (clone: 53-6.7, catalog number: 553034, 1:100 dilution) were purchased from BD Biosciences. Anti-mouse CD4-PerCP/Cy5.5 (clone: GK1.5, catalog number: 100434, 1:100 dilution), anti-mouse CD8a-Pacific Blue (clone: 53-6.7, catalog number: 100725, 1:100 dilution), anti-mouse CD3 ϵ -BV605 (clone: 145-2C11, catalog number: 100351, 1:100 dilution), anti-mouse IgD-BV711 (clone: 11-26c.2a, catalog number: 405731, 1:100 dilution), anti-mouse/human CD45R/B220-BV785 (clone: RA3-6B2, catalog number: 103246, 1:100 dilution), anti-mouse CD273-PE (clone: TY25, catalog number: 107206, 1:100 dilution), anti-mouse CD4-PE/Cy5 (clone: H129.19, catalog number: 130312, 1:100 dilution), anti-mouse Ly-6G/Ly-6C (Gr-1)-PE/Cy5 (clone: RB6-8C5, catalog number: 108410, 1:100 dilution), anti-mouse/human GL7-PE/Cy7 (clone: GL7, catalog number: 144620, 1:100 dilution) were purchased from BioLegend. Anti-mouse CD38-AF700 (catalog number: 56-0381-82, 1:100 dilution), anti-mouse F4/80-PE-Cy5 (clone: BM8, catalog number: 15-4801-82, 1:100 dilution), anti-mouse CD19-PE/Cy5.5 (clone: 1D3 catalog number: 35-0193-82, 1:100 dilution) were purchased from ThermoFisher. Mouse anti-VSV Indiana G antibody (clone: 8G5F11, catalog number: Ab01401-2.0, 1:100 dilution) was supplied by Absolute Antibody. InVivoMAb anti-mouse CD28 (clone: D665, catalog number: BE0328) was purchased from BioXcell.

Validation

All antibodies were verified by the supplier and each lot has been quality tested. All the antibodies used are from commercial sources and have been validated by the vendors. Validation data are available on the manufacturer's website.

1. Rat anti-Mouse CD107a-FITC antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/fic-rat-anti-mouse-cd107a.553793>)
2. Rat anti-mouse IL-4-BV786 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv786-rat-anti-mouse-il-4.564006>)
3. Rat anti-mouse IL-2-APC antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/apc-rat-anti-mouse-il-2.554429>)
4. Rat anti-mouse IFN- γ -AF700 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/alexa-fluor-700-rat-anti-mouse-ifn.557998>)
5. Rat anti-mouse TNF-PE/Cy7 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/pe-cy-7-rat-anti-mouse-tnf.557644>)
6. Hamster anti-mouse CD80-BV650 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/bv650-hamster-anti-mouse-cd80.563687>)
7. Rat anti-mouse IgM-PE/CF594 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://wwwbdbiosciences.com/en-us/products/reagents/flow-cytometry-reagents/research-reagents/single-color-antibodies-ruo/pe-cf594-rat-anti-mouse-igm.562565>)
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9. Anti-mouse CD4-PerCP/Cy5.5 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/percp-cyanine5-5-anti-mouse-cd4-antibody-4220>)
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11. Anti-mouse CD3 ϵ -BV605 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/brilliant-violet-605-anti-mouse-cd3epsilon-antibody-11974>)
12. Anti-mouse IgD-BV711 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/brilliant-violet-711-anti-mouse-igd-9572>)
13. Anti-mouse/human CD45R/B220-BV785 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/brilliant-violet-785-anti-mouse-human-cd45r-b220-antibody-7960>)
14. Anti-mouse CD273-PE antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/pe-anti-mouse-cd273-b7-dc-pd-l2-antibody-2547>)
15. Anti-mouse CD4-PE/Cy5 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/pe-cyanine5-anti-mouse-cd4-antibody-5489>)
16. Anti-mouse Ly-6G/Ly-6C (Gr-1)-PE/Cy5 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/pe-cyanine5-anti-mouse-ly-6g-ly-6c-gr-1-antibody-461>)
17. Anti-mouse/human GL7-PE/Cy7 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.biolegend.com/en-us/products/pe-cyanine7-anti-mouse-human-gl7-antigen-t-and-b-cell-activation-marker-antibody-17249>)
18. Anti-mouse CD38-AF700 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.thermofisher.com/antibody/product/CD38-Antibody-clone-90-Monoclonal/56-0381-82>)
19. Anti-mouse F4/80-PE-Cy5 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.thermofisher.com/antibody/product/F4-80-Antibody-clone-BM8-Monoclonal/15-4801-82>)
20. Anti-mouse CD19-PE/Cy5.5 antibody has been validated to be used for flow analysis and mentioned species reactivity with mouse. (<https://www.thermofisher.com/antibody/product/CD19-Antibody-clone-eBio1D3-1D3-Monoclonal/35-0193-82>)
21. Anti-mouse CD28 antibody has been validated to be used for T cell activation and mentioned species reactivity with mouse. (https://bioxccl.com/invivomab-anti-mouse-cd28-be0328?queryID=a65a1fbde10cc6c5a9783e047db32ac4&objectID=30748&indexName=bioxccl_live_default_products)

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	The murine colon cancer cell lines MC38 and B16F10 was obtained from from ATCC, A549-ACE2 cells was a gift from Prof. Susan Weiss (Penn medicine), all of the cell lines were tested negative for mycoplasma in University of Pennsylvania cell center.
Authentication	A short tandem repeat DNA profiling method was used to authenticate the cell lines and the results were compared with reference database. Moreover, no mycoplasma contamination was detected in the above cell lines.
Mycoplasma contamination	All cell lines were tested for mycoplasma contamination. No mycoplasma contamination was found.
Commonly misidentified lines (See ICLAC register)	MC38 and A549-ACE2 cell lines cell lines are not listed in the database.

Animals and other research organisms

Policy information about [studies involving animals; ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	C57BL/6 mice and Ai9 mice (female, 6-8 weeks) were ordered from Jackson laboratory and housed in a specific-pathogen-free animal facility at ambient temperature (22 ± 2 °C), air humidity 40%–70% and 12-h dark/12-h light cycle.
Wild animals	No wild animal was used in this study.
Reporting on sex	Female mice were used in this study
Field-collected samples	No field-collected sample was used in this study.
Ethics oversight	All animal experiment protocols were reviewed and approved by the institutional animal care and use committee of the University of Pennsylvania.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Cells was incubated with ammonium chloride buffer for erythrocyte lysis and washed with PBS. Single-cell suspensions were obtained by filtering through a 70 μ M cell strainer. Then the suspensions and stained with antibodies according to the manufacturer's protocols, and then analyzed by flow cytometry.
Instrument	BD LSR II

Software	FlowJo software package (Flowjo V10)
Cell population abundance	The absolute cells around 8000-10000 were analyzed for fluorescent intensity in the defined gate.
Gating strategy	In general, cells were first gated on FSC/SSC. Singlet cells were gated using FSC-H and FSC-A. Dead cells were then excluded and further surface and intracellular antigen gating was performed on the live cell population. Gate strategies are provided in the Supplementary information of this paper

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.